C h a p t e r 4

Full-Wave Rectifiers

Converting ac to dc

4.1 introduction

The objective of a full-wave rectifier is to produce a voltage or current that is purely dc or has some specified DC component. While the purpose of the full-wave rectifier is basically the same as that of the half-wave rectifier, full-wave rectifiers have some fundamental advantages. The average current in the ac source is zero in the full-wave rectifier, thus avoiding problems associated with nonzero average source currents, particularly in transformers. The output of the full-wave rectifier has inherently less ripple than the half-wave rectifier.

In this chapter, uncontrolled and controlled single-phase and three-phase full-wave converters used as rectifiers are analyzed for various types of loads.

Also included are examples of controlled converters operating as inverters, where power flow is from the DC side to the AC side.

4.2 Single-Phase Full-Wave Rectifiers

The bridge rectifier and the center-tapped transformer rectifier of Figs. 4-1 and 4-2 are two basic single-phase full-wave rectifiers.

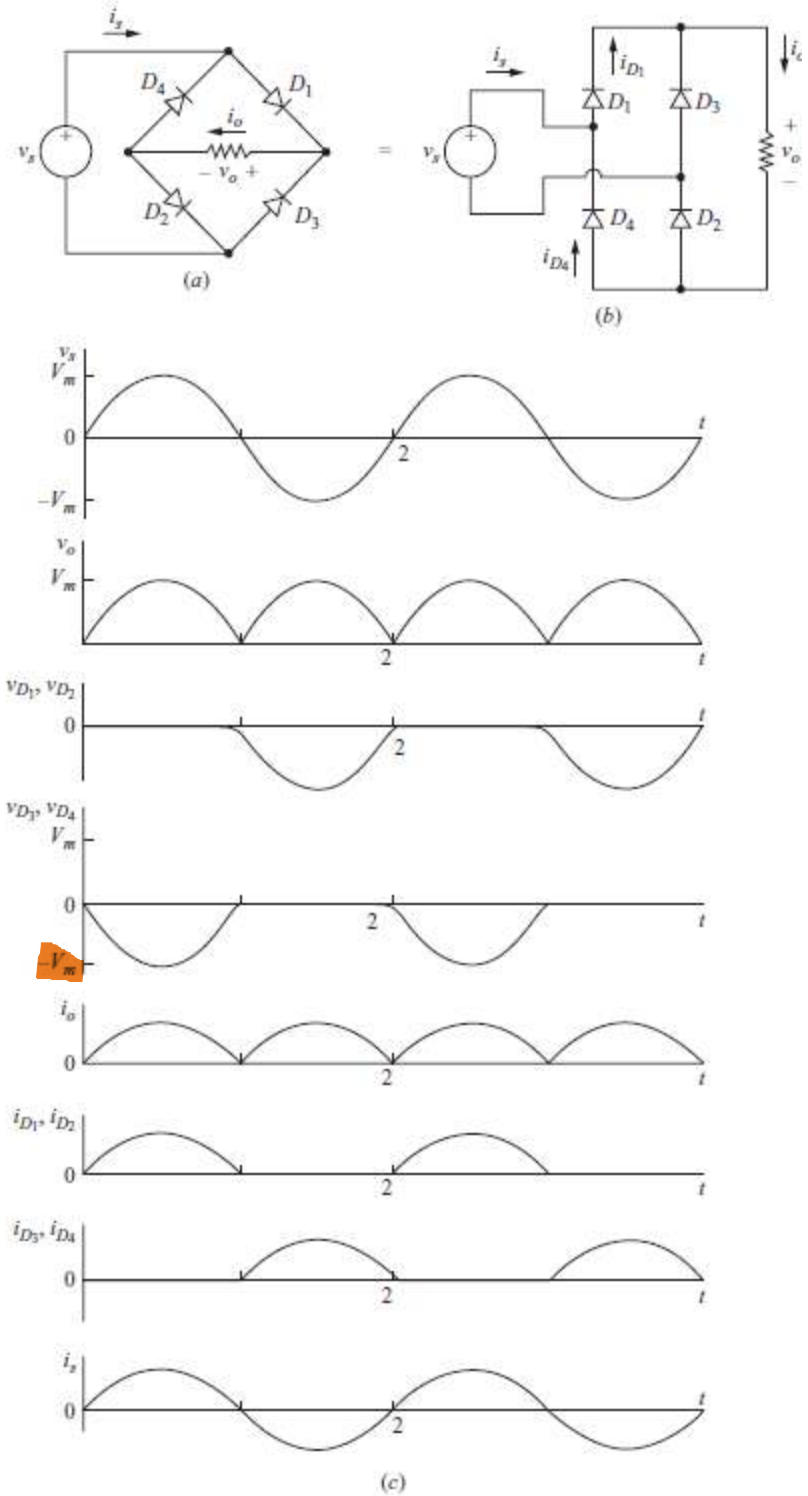


Figure 4-1 Full-wave bridge rectifier. (*a*) Circuit diagram. (*b*) Alternative representation. (*c*) Voltages and currents.

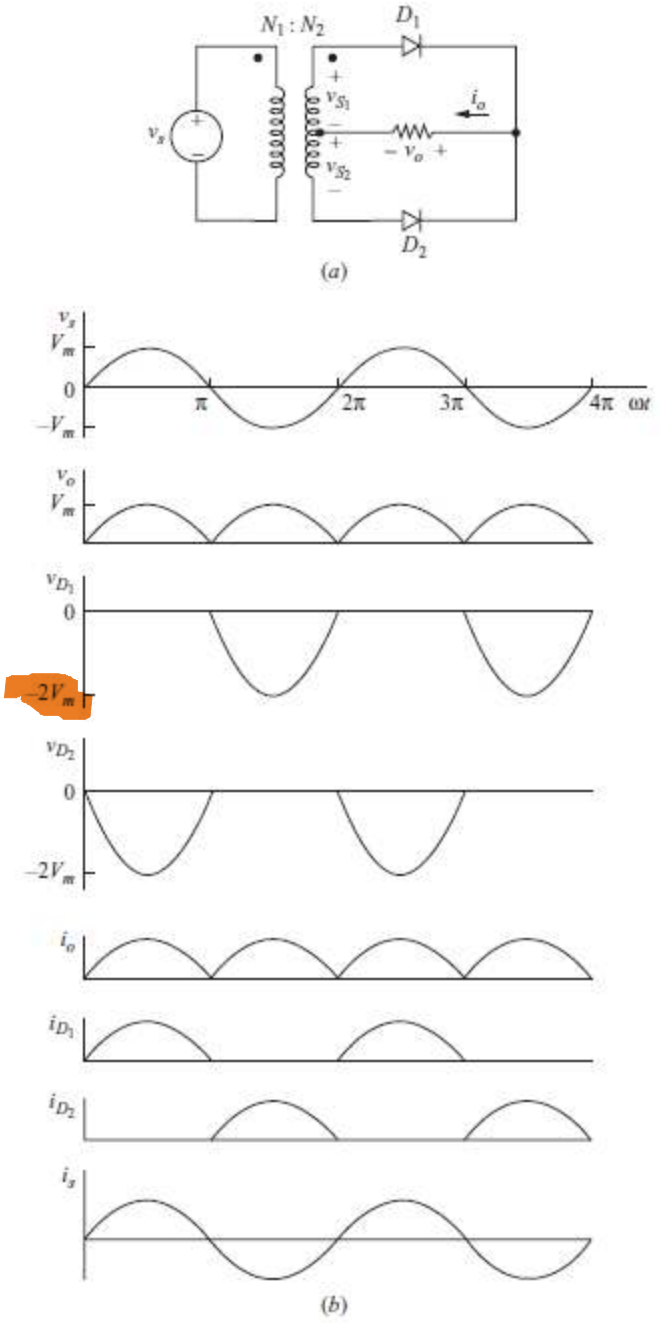


Figure 4-2 Full-wave center-tapped rectifier (*a*) circuit; (*b*) voltages and currents.

The Bridge Rectifier

For the bridge rectifier of Fig. 4-1, these are some basic observations:

1. Diodes D_1 and D_2 conduct together, and D_3 and D_4 conduct together. Kirchhoff's voltage law around the loop containing the source, D_1, and D_3 cannot conduct simultaneously. The load current can be positive or zero but can never be negative.

2. The voltage across the load is $+v_S$ when D_1 and D_2 are on. The voltage across the load is $-v_S$ when D_3 and D_4 are on.
3. The maximum voltage across a reverse-biased diode is the peak value of the source. This can be shown by Kirchhoff's voltage law around the loop containing the source, D_1, and D_3. With D_1 on, the voltage across D_3 is$-v_S$.
4. The current entering the bridge from the source is $i_{D1} - i_{D4}$, which is symmetric about zero. Therefore, the average source current is zero.
5. The rms source current is the same as the rms load current. The source current is the same as the load current for one-half of the source period and is the negative of the load current for the other half. The squares of the load and source currents are the same, so the rms currents are equal.
6. The fundamental frequency of the output voltage is 2 ω, where ω is the frequency of the AC input since two periods of the output occur for every period of the input. The Fourier series of the output consists of a DC term and the even harmonics of the source frequency.

The Center-Tapped Transformer Rectifier

The voltage waveforms for a resistive load in the rectifier using a center-tapped transformer are shown in Fig. 4-2. Some basic observations for this circuit are as follows:
1. Kirchhoff's voltage law shows that only one diode can conduct at a time. The load current can be positive or zero but never negative.
2. The output voltage is $+v_{S1}$ when D_1 conducts and is $-v_{S2}$ when D_2 conducts.
The transformer secondary voltages are related to the source voltage by $v_{S1} = v_{S2} = v_S(N_2/2\,N_1)$.

3. Kirchhoff's voltage law around the transformer secondary windings, D_1, and D_2 shows that the maximum voltage across a reverse-biased diode is *twice* the peak value of the load voltage.
4. Current in each half of the transformer secondary is reflected to the primary, resulting in an average source current of zero.
5. The transformer provides electrical isolation between the source and the load.
6. The fundamental frequency of the output voltage is 2 ω since two periods of the output occur for every period of the input.
The lower peak diode voltage in the bridge rectifier makes it more suitable for high-voltage applications. The center-tapped transformer rectifier, in addition to including electrical isolation, has only one diode voltage drop

between the source and load, making it desirable for low-voltage, high-current applications.
The following discussion focuses on the full-wave bridge rectifier but generally applies to the center-tapped circuit as well.

Resistive Load

The voltage across a resistive load for the bridge rectifier of Fig. 4-1 is expressed as:

$$v_o(\omega t) = \begin{cases} V_m \sin \omega t & \text{for } 0 \le \omega t \le \pi \\ -V_m \sin \omega t & \text{for } \pi \le \omega t \le 2\pi \end{cases} \tag{4-1}$$

The dc component of the output voltage is the average value, and load current is simply the resistor voltage divided by resistance.

$$V_o = \frac{1}{\pi}\int_0^{\pi} V_m \sin \omega t \, d(\omega t) = \frac{2V_m}{\pi}$$

$$I_o = \frac{V_o}{R} = \frac{2V_m}{\pi R} \tag{4-2}$$

Power absorbed by the load resistor can be determined from $I_{rms}^2 R$, where I_{rms} for the full-wave rectified current waveform is the same as for an unrectified sine wave,

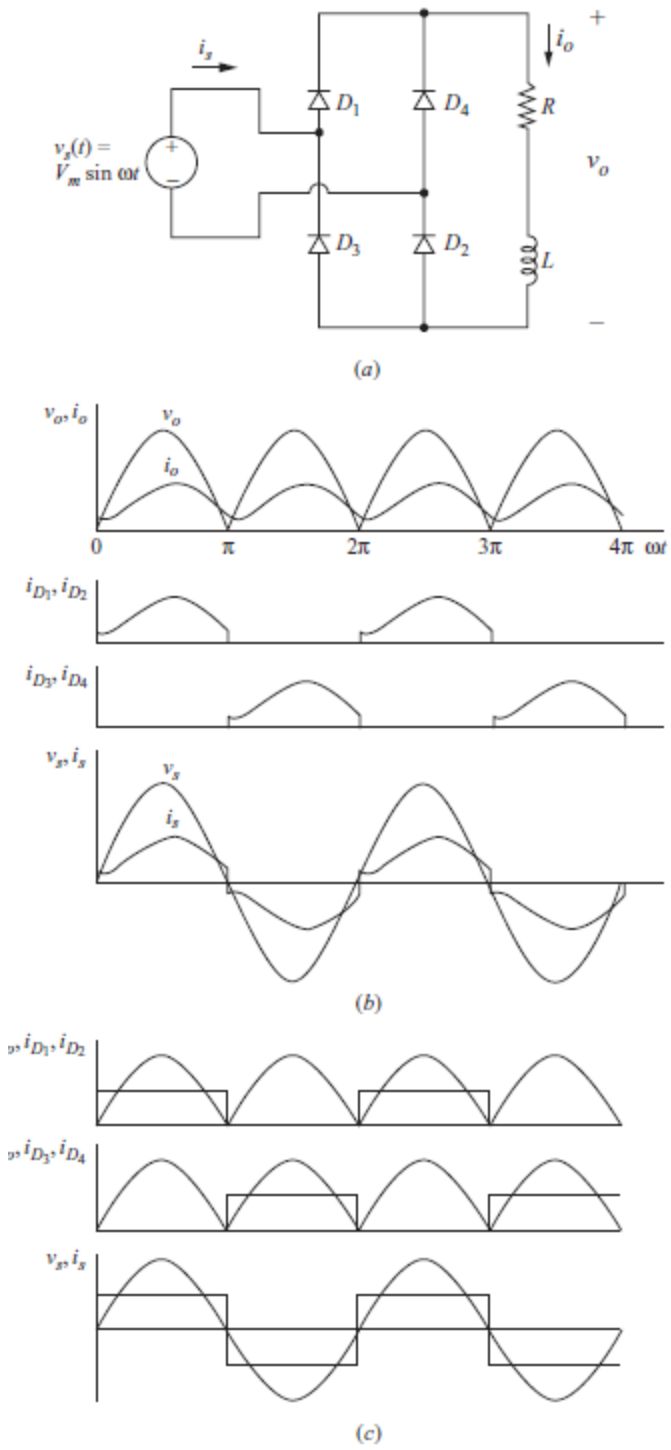
$$I_{\text{rms}} = \frac{I_m}{\sqrt{2}} \tag{4-3}$$

The source current for the full-wave rectifier with a resistive load is a sinusoid that is in phase with the voltage, so the power factor is 1.

***RL* Load**

For an *RL* series-connected load (Fig. 4-3*a*), the method of analysis is similar to that for the half-wave rectifier with the freewheeling diode discussed in Chap. 3.

After a transient that occurs during start-up, the load current i_o reaches a periodic steady-state condition similar to that in Fig. 4-3*b*.



لا يوجد لعدم وجودFWD
زمن لتفريغ التيار من الملف

Figure 4-3 (*a*) Bridge rectifier with an *RL* load; (*b*) Voltages and currents; (*c*) Diode and source currents when the inductance is large and the current is nearly constant.

For the bridge circuit, current is transferred from one pair of diodes to the other pair when the source changes polarity. The voltage across the *RL* load is a full-wave rectified sinusoid, as it was for the resistive load.

The full-wave rectified sinusoidal voltage across the load can be expressed as a Fourier series consisting of a dc term and the even harmonics

$$v_o(t) = V_o + \sum_{n=2,4\ldots}^{\infty} V_n \cos(n\omega_0 t + \pi)$$

where (4-4)

$$V_o = \frac{2V_m}{\pi} \quad \text{and} \quad V_n = \frac{2V_m}{\pi}\left(\frac{1}{n-1} - \frac{1}{n+1}\right)$$

The current in the *RL* load is then computed using superposition, taking each frequency separately and combining the results. The DC current and current amplitude at each frequency are computed from:

$$I_0 = \frac{V_0}{R}$$

$$I_n = \frac{V_n}{Z_n} = \frac{V_n}{|R + jn\omega L|} \qquad (4\text{-}5)$$

Note that as the harmonic number *n* increases in Eq.(4-4), the voltage amplitude decreases. For an *RL* load, the impedance Z_n increases as *n* increases.

The combination of decreasing V_n and increasing Z_n makes $I_n$decrease rapidly for increasing harmonic number. Therefore, the dc term and only a few, if any, of the AC terms are usually necessary to describe current in an *RL* load.

EXAMPLE 4-1(Full-Wave Rectifier with *RL* Load)

The bridge rectifier circuit of Fig. 4-3*a* has an ac source with $V_m = 100\ V$ at 60 Hz and a series *RL* load with $R = 10\ \Omega$ and $L = 10\ mH$.

(*a*) Determine the average current in the load.

(*b*) Estimate the peak-to-peak variation in load current based on the first AC term in the Fourier series.

(*c*) Determine the power absorbed by the load and the power factor of the circuit.

(*d*) Determine the average and rms currents in the diodes.

■ **Solution**

(*a*) The average load current is determined from the dc term in the Fourier series. The voltage across the load is a full-wave rectified sine wave that has the Fourier series determined from Eq. (4-4). Average output voltage is:

$$V_0 = \frac{2V_m}{\pi} = \frac{2(200)}{\pi} = 63.7 \text{ V}$$

and average load current is

$$I_0 = \frac{V_0}{R} = \frac{63.7 \text{ V}}{10\ \Omega} = 6.37 \text{ A}$$

(b) Amplitudes of the ac voltage terms are determined from Eq. (4-4). For $n = 2$ and 4,

$$V_2 = \frac{2(100)}{\pi}\left(\frac{1}{1} - \frac{1}{3}\right) = 42.4 \text{ V}$$

$$V_4 = \frac{2(100)}{\pi}\left(\frac{1}{3} - \frac{1}{5}\right) = 8.49 \text{ V}$$

The amplitudes of first two AC current terms in the current Fourier series are computed from Eq. (4-5).

$$I_2 = \frac{42.4}{|10 + j(2)(377)(0.01)|} = \frac{42.4 \text{ V}}{12.5\ \Omega} = 3.39 \text{ A}$$

$$I_4 = \frac{8.49}{|10 + j(4)(377)(0.01)|} = \frac{8.49 \text{ V}}{18.1\ \Omega} = 0.47 \text{ A}$$

The current I_2 is much larger than I_4 and higher-order harmonics, so I_2 can be used to estimate the peak-to-peak variation in load current $\Delta i_o \approx 2(3.39) = 6.78\ A$.
Actual variation in i_o will be larger because of the higher-order terms.

(c) The power absorbed by the load is determined from I_{rms}^2. The rms current is then determined from Eq. (2-43) as:

$$I_{\text{rms}} = \sqrt{\sum I_{n,\text{rms}}^2}$$

$$= \sqrt{(6.37)^2 + \left(\frac{3.39}{\sqrt{2}}\right)^2 + \left(\frac{0.47}{\sqrt{2}}\right)^2 + \cdots} \approx 6.81 \text{ A}$$

Adding more terms in the series would not be useful because they are small and have little effect on the result. Power in the load is:

$$P = I_{\text{rms}}^2 R = (6.81)^2(10) = 464 \text{ W}$$

The *rms* source current is the same as the *rms* load current. Power factor is:

$$\text{pf} = \frac{P}{S} = \frac{P}{V_{s,\text{rms}} I_{s,\text{rms}}} = \frac{464}{\left(\frac{100}{\sqrt{2}}\right)(6.81)} = 0.964$$

(d) Each diode conducts for one-half of the time, so

$$I_{D,\text{avg}} = \frac{I_o}{2} = \frac{6.37}{2} = 3.19 \text{ A}$$

and

$$I_{D,\text{rms}} = \frac{I_{\text{rms}}}{\sqrt{2}} = \frac{6.81}{\sqrt{2}} = 4.82 \text{ A}$$

In some applications, the load inductance may be relatively large or made large by adding external inductance. If the inductive impedance for the ac terms in the Fourier series effectively eliminates the AC current terms in the load, the load current is essentially DC. If $\omega L \gg R$,

$$i(\omega t) \approx I_o = \frac{V_o}{R} = \frac{2V_m}{\pi R} \qquad \text{for } \omega L \gg R \qquad (4\text{-}6)$$

$$I_{\text{rms}} \approx I_o$$

Load and source voltages and currents are shown in Fig. 4-3*c*.

Source Harmonics

Nonsinusoidal source current is a concern in power systems. Source currents like that of Fig. 4-3 have a fundamental frequency equal to that of the source but are rich in the odd-numbered harmonics. Measures such as total harmonic distortion (THD) and distortion factor (DF) as presented in Chap. 2 describe the nonsinusoidal property of the source current. Where harmonics are of concern, filters can be added to the input of the rectifier.

***RL*-Source Load**

Another general industrial load may be modeled as a series resistance, inductance, and a DC voltage source, as shown in Fig. 4-5*a*. A DC motor drive circuit and a battery charger are applications for this model. There are two possible modes of operation for this circuit, the continuous-current mode and the discontinuous-current mode. In the continuous-current mode,

the load current is always positive for steady-state operation (Fig. 4-5*b*). Discontinuous load current is characterized by current returning to zero during every period (Fig. 4-5*c*).

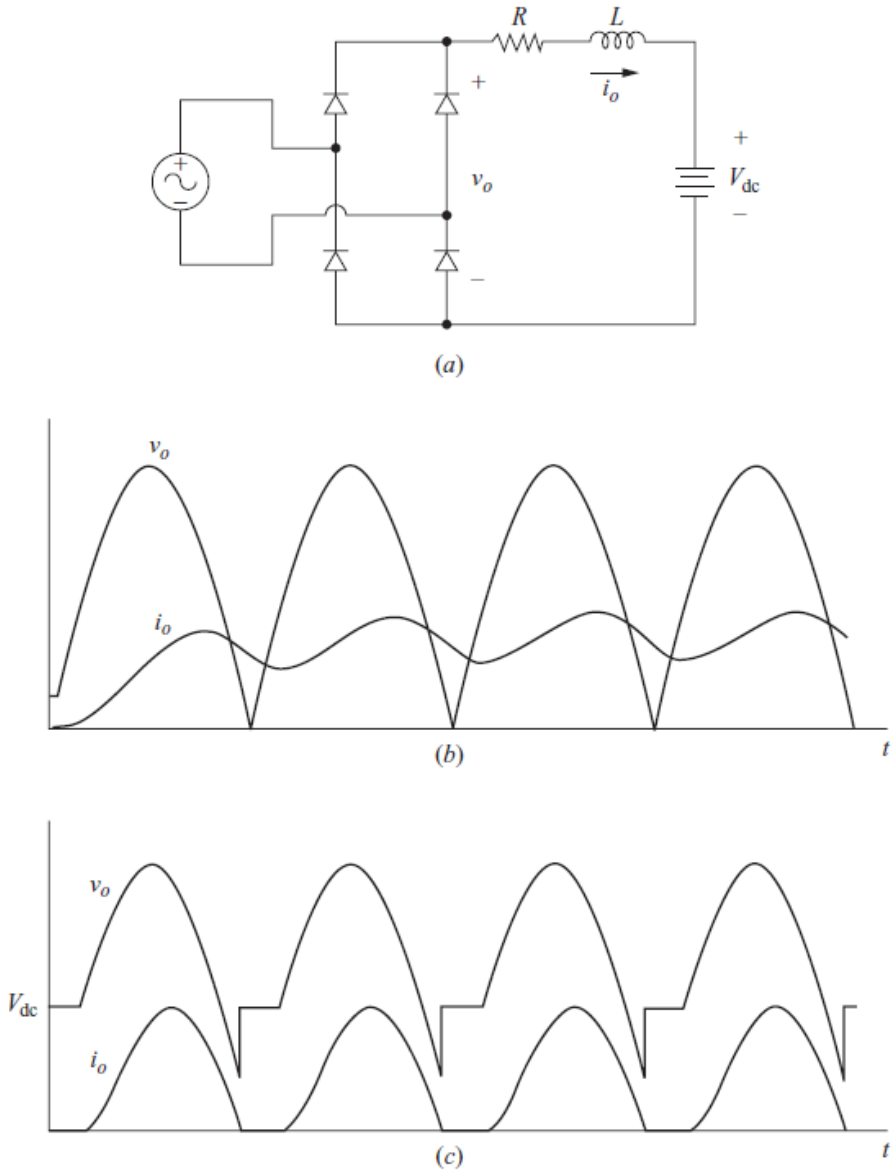


Figure 4-5 (*a*) Rectifier with *RL*-source load; (*b*) Continuous current: when the circuit is energized, the load current reaches the steady-state after a few periods; (*c*) Discontinuous current: the load current returns to zero during every period.

For continuous-current operation, one pair of diodes is always conducting, and the voltage across the load is a full-wave rectified sine wave. The only modification to the analysis that was done for an *RL* load is in the DC term of the Fourier series. The dc (average) component of current in this circuit is:

$$I_o = \frac{V_o - V_{dc}}{R} = \frac{\frac{2V_m}{\pi} - V_{dc}}{R} \qquad (4\text{-}7)$$

The sinusoidal terms in the Fourier analysis are unchanged by the dc source provided that the current is continuous.
Discontinuous current is analyzed like the half-wave rectifier of Sec. 3.5. The load voltage is not a full-wave rectified sine wave for this case, so the Fourier series of Eq. (4-4) does not apply.

Example 4-3(Full-Wave Rectifier with *RL*-Source Load—Continuous Current)
For the full-wave bridge rectifier circuit of Fig. 4-5*a*, the AC source is $120\ Vrms\ at\ 60\ Hz, R = 2\ \Omega, L = 10\ mH$, and $V_{dc} = 80\ V$. Determine the power absorbed by the DC voltage source and the power absorbed by the load resistor.

■ **Solution**

For continuous current, the voltage across the load is a full-wave rectified sine wave which has the Fourier series given by Eq. (4-4). Equation (4-7) is used to compute the average current, which is used to compute power absorbed by the dc source,

$$I_0 = \frac{\frac{2V_m}{\pi} - V_{dc}}{R} = \frac{\frac{2\sqrt{2}(120)}{\pi} - 80}{2} = 14.0\ \text{A}$$

$$P_{dc} = I_0 V_{dc} = (14)(80) = 1120\ \text{W}$$

The first few terms of the Fourier series using Eqs. (4-4) and (4-5) are shown in Table 4-1.

Table 4-1 Fourier series components

n	V_n	Z_n	I_n
0	108	2.0	14.0
2	72.0	7.80	9.23
4	14.4	15.2	0.90

The rms current is computed from Eq. (2-43).

$$I_{rms} = \sqrt{14^2 + \left(\frac{9.23}{\sqrt{2}}\right)^2 + \left(\frac{0.90}{\sqrt{2}}\right)^2 + \cdots} \approx 15.46\ \text{A}$$

Power absorbed by the resistor is

$$P_R = I_{\text{rms}}^2 R = (15.46)^2(2) = 478 \text{ W}$$

Capacitance Output Filter

Placing a large capacitor in parallel with a resistive load can produce an output voltage that is essentially DC (Fig. 4-6).

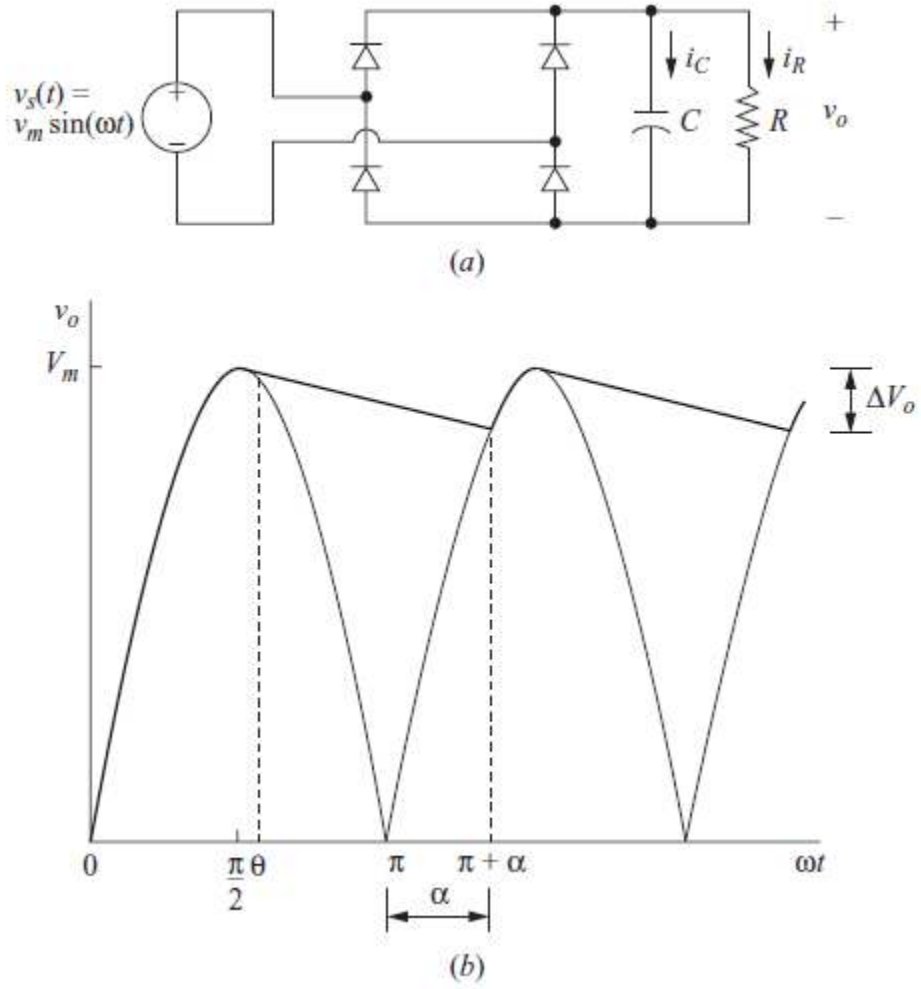


Figure 4-6 (*a*) Full-wave rectifier with capacitance filter; (*b*) Source and output voltage.

The analysis is very much like that of the half-wave rectifier with a capacitance filter in Chap. 3. In the full-wave circuit, the time that the capacitor discharges is smaller than that for the half-wave circuit because of the rectified sine wave in the second half of each period. The output voltage ripple for the full-wave rectifier is approximately one-half that of the half-

wave rectifier. The peak output voltage will be less in the full-wave circuit because there are two diode voltage drops rather than one.
The analysis proceeds exactly as for the half-wave rectifier. The output voltage is a positive sine function when one of the diode pairs is conducting and is a decaying exponential otherwise. Assuming ideal diodes,

$$v_o(\omega t) = \begin{cases} |V_m \sin \omega t| & \text{one diode pair on} \\ (V_m \sin \theta) e^{-(\omega t - \theta)/\omega RC} & \text{diodes off} \end{cases} \tag{4-8}$$

where θ is the angle where the diodes become reverse biased, which is the same as that for the half-wave rectifier and is found using Eq. (3-41).

$$\theta = \tan^{-1}(-\omega RC) = -\tan^{-1}(\omega RC) + \pi \tag{4-9}$$

The maximum output voltage is V_m, and the minimum output voltage is determined by evaluating v_o at the angle at which the second pair of diodes turns on, which is at $\omega t = \pi + \alpha$. At that boundary point,

$$(V_m \sin \theta) e^{-(\pi + \alpha - \theta)/\omega RC} = -V_m \sin(\pi + \alpha)$$

Or

$$(\sin \theta) e^{-(\pi + \alpha - \theta)/\omega RC} - \sin \alpha = 0 \tag{4-10}$$

Which must be solved numerically for α.
The peak-to-peak voltage variation, or ripple, is the difference between the maximum and minimum voltages.

$$\boxed{\Delta V_o = V_m - |V_m \sin(\pi + \alpha)| = V_m(1 - \sin \alpha)} \tag{4-11}$$

This is the same as Eq. (3-49) for voltage variation in the half-wave rectifier, but α is larger for the full-wave rectifier and the ripple is smaller for a given load. Capacitor current is described by the same equations as for the half-wave rectifier.
In practical circuits where $\omega RC \gg \pi$.

$$\theta \approx \pi/2 \qquad \alpha \approx \pi/2 \tag{4-12}$$

The minimum output voltage is then approximated from Eq. (4-9) for the diodes off evaluated at $\omega t = \pi$.

$$v_o(\pi + \alpha) = V_m e^{-(\pi + \pi/2 - \pi/2)/\omega RC} = V_m e^{-\pi/\omega RC}$$

The ripple voltage for the full-wave rectifier with a capacitor filter can then be approximated as

$$\Delta V_o \approx V_m(1 - e^{-\pi/\omega RC})$$

Furthermore, the exponential in the above equation can be approximated by the series expansion

$$e^{-\pi/\omega RC} \approx 1 - \frac{\pi}{\omega RC}$$

Substituting for the exponential in the approximation, the peak-to-peak ripple is

$$\boxed{\Delta V_o \approx \frac{V_m \pi}{\omega RC} = \frac{V_m}{2fRC}} \qquad (4\text{-}13)$$

Note that the approximate peak-to-peak ripple voltage for the full-wave rectifier is one-half that of the half-wave rectifier from Eq. (3-51). As for the half-wave rectifier, the peak diode current is much larger than the average diode current and Eq. (3-48) applies. The average source current is zero.

Example 4-4(Full-Wave Rectifier with Capacitance Filter)
The full-wave rectifier of Fig. 4-6*a* has a $120\ Vsource$ at 60 Hz, $R = 500\ \Omega, and\ C = 100\ \mu F$.
(*a*) Determine the peak-to-peak voltage variation of the output. (*b*) Determine the value of capacitance that would reduce the output voltage ripple to 1 percent of the DC value.

■ Solution

From the parameters given,

$$V_m = 120\sqrt{2} = 169.7\ \text{V}$$

$$\omega RC = (2\pi 60)(500)(10)^{-6} = 18.85$$

The angle θ is determined from Eq. (4-9).

$$\theta = -\tan^{-1}(18.85) + \pi = 1.62\ \text{rad} = 93°$$

$$V_m \sin\theta = 169.5\ \text{V}$$

The angle α is determined by the numerical solution of Eq. (4-10).

$$\sin(1.62)e^{-(\pi+\alpha-1.62)/18.85} - \sin\alpha = 0$$

$$\alpha = 1.06 \text{ rad} = 60.6°$$

(a) Peak-to-peak output voltage is described by Eq. (4-11).

$$\Delta V_o = V_m(1 - \sin\alpha) = 169.7[1 - \sin(1.06)] = 22 \text{ V}$$

Note that this is the same load and source as for the half-wave rectifier of Example 3-9 where $\Delta Vo = 43\,V$.

(b) With the ripple limited to 1 percent, the output voltage will be held close to V_m and the approximation of Eq. (4-13) applies.

$$\frac{\Delta V_o}{V_m} = 0.01 \approx \frac{1}{2fRC}$$

Solving for *C*,

$$C \approx \frac{1}{2fR(\Delta V_o/V_m)} = \frac{1}{(2)(60)(500)(0.01)} = 1670\ \mu\text{F}$$

Voltage Doublers

The rectifier circuit of Fig. 4-7*a* serves as a simple voltage doubler, having an output of twice the peak value of the source. For ideal diodes C_1 charges to V_m through D_1 when the source is positive; C_2 charges to V_m through D_2 when the source is negative. The voltage across the load resistor is the sum of the capacitor voltages $2\,V_m$. This circuit is useful when the output voltage of a rectifier must be larger than the peak input voltage. Voltage doubler circuits avoid using a transformer to step up the voltage, saving expense, volume, and weight.

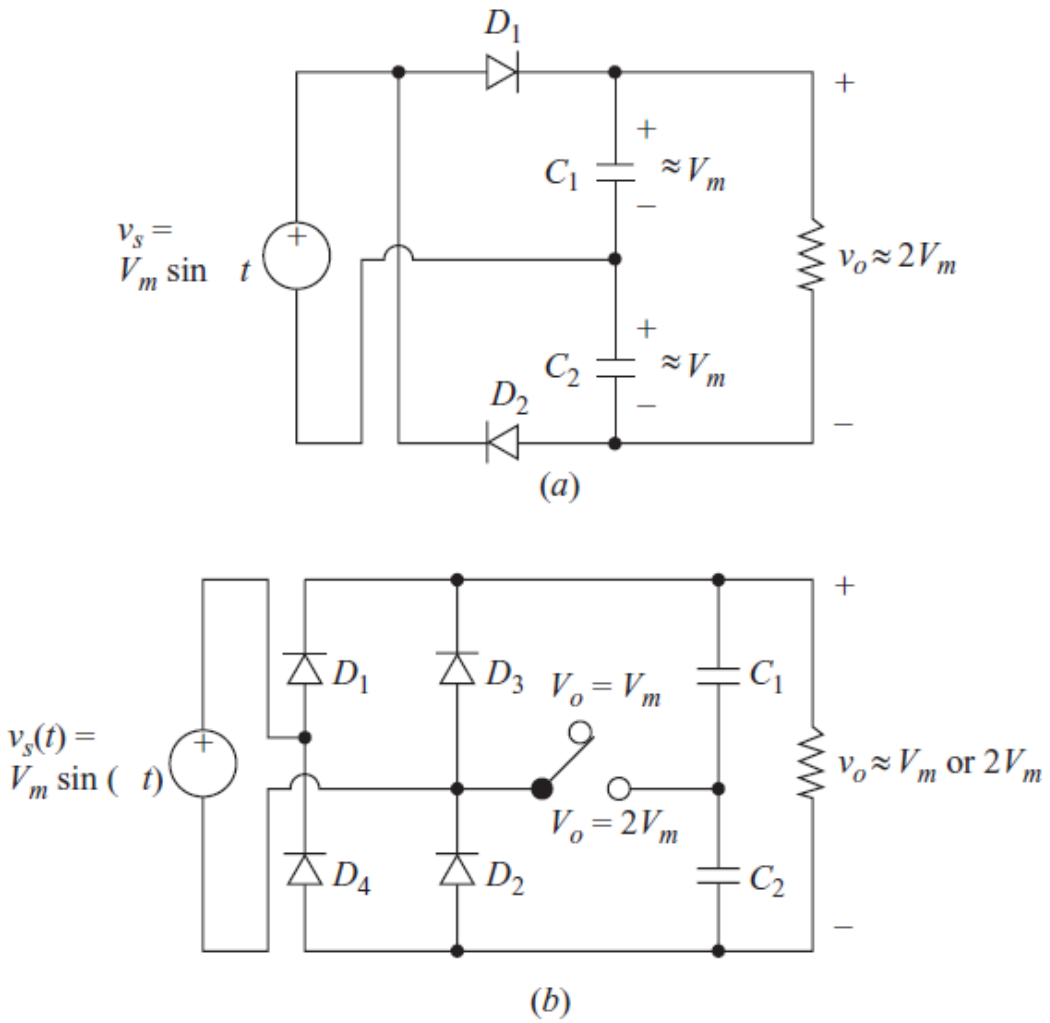


Figure 4-7 (*a*) Voltage doubler. (*b*) Dual-voltage rectifier.

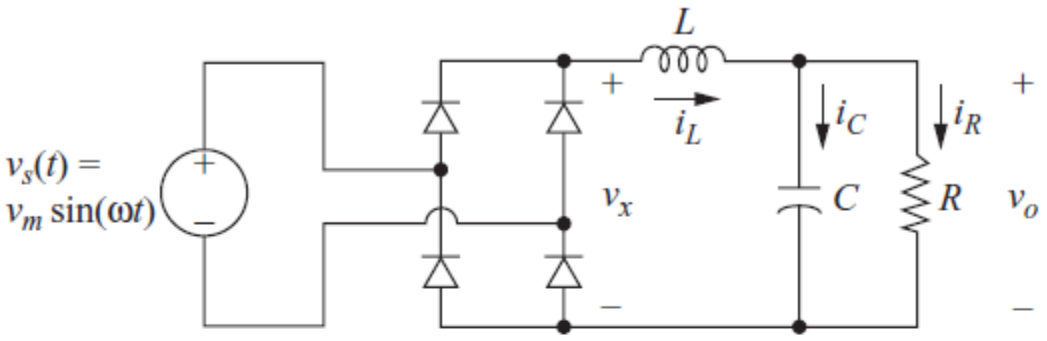
The full-wave rectifier with a capacitive output filter can be combined with the voltage doubler, as shown in Fig. 4-7*b*. When the switch is open, the circuit is similar to the full-wave rectifier of Fig. 4-6*a*, with output at approximately V_m when the capacitors are large. When the switch is closed, the circuit acts as the voltage doubler of Fig. 4-7*a*. Capacitor C_1 charges to V_m through D_1 when the source is positive, and C_1 charges to V_m through D_4 when the source is negative.

The output voltage is then 2 V_m. Diodes D_2 and D_3 remain reverse-biased in this mode.

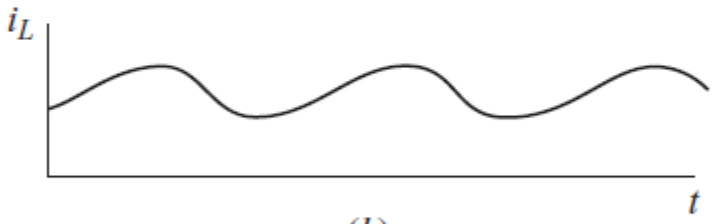
This voltage doubler circuit is useful when equipment must be used on systems with different voltage standards. For example, a circuit could be designed to operate properly in both the United States, where the line voltage is 120 V, and places abroad where the line voltage is 240 V.

LC Filtered Output

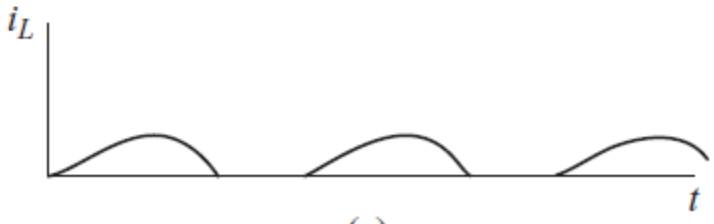
Another full-wave rectifier configuration has an *LC* filter on the output, as shown in Fig. 4-8*a*. The purpose of the filter is to produce an output voltage that is close to purely DC. The capacitor holds the output voltage at a constant level, and the inductor smooths the current from the rectifier and reduces the peak current in the diodes from that of the current of Fig. 4-6*a*. The circuit can operate in the continuous- or discontinuous-current mode. For continuous current, the inductor current is always positive, as illustrated in Fig. 4-8*b*. Discontinuous current is characterized by the inductor current returning to zero in each cycle, as illustrated in Fig. 4-8*c*. The continuous-current case is easier to analyze and is considered first.



(*a*)



(*b*)



(*c*)

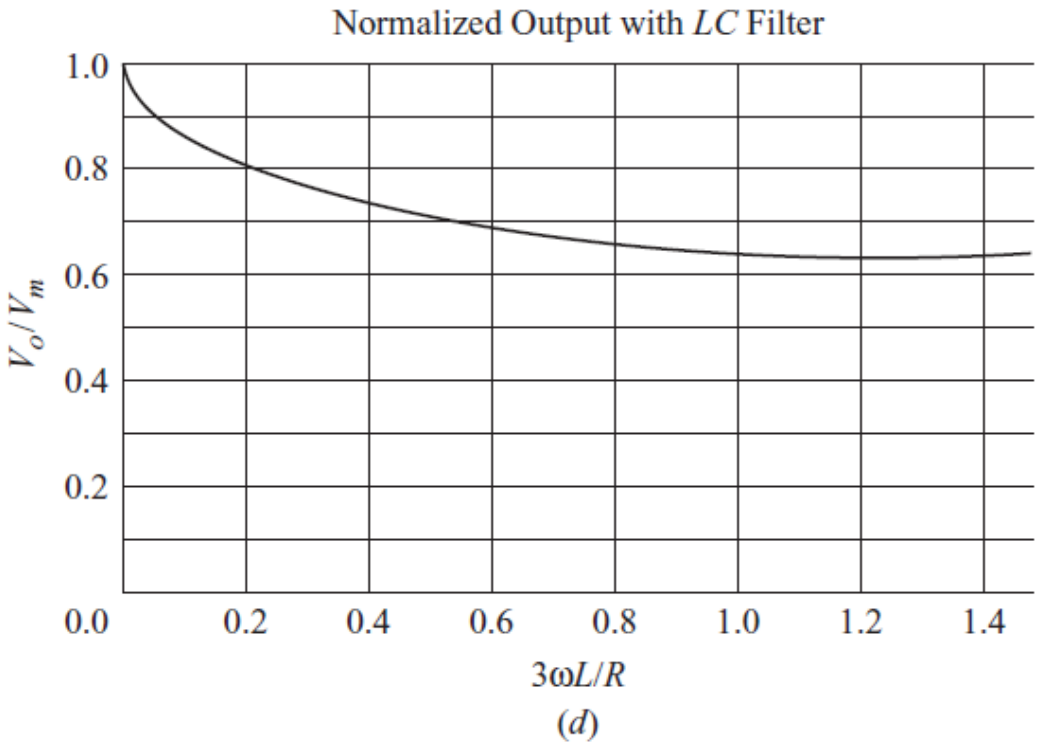


Figure 4-8 (*a*) Rectifier with *LC* filtered output; (*b*) Continuous inductor current; (*c*) Discontinuous inductor current; (*d*) Normalized output.

Continuous Current for *LC* Filtered Output

For continuous current, the voltage v_x in Fig. 4-8*a* is a full-wave rectified sine wave, which has an average value of $2V_m/\pi$. Since the average voltage across the inductor in the steady state is zero, the average output voltage for continuous inductor current is:

$$V_o = \frac{2V_m}{\pi} \qquad (4\text{-}14)$$

Average inductor current must equal the average resistor current because the average capacitor current is zero.

$$I_L = I_R = \frac{V_o}{R} = \frac{2V_m}{\pi R} \qquad (4\text{-}15)$$

The variation in inductor current can be estimated from the first ac term in the Fourier series. The first ac voltage term is obtained from Eq. (4-4) with $n = 2$.

Assuming the capacitor to be a short circuit to ac terms, the harmonic voltage v_2 exists across the inductor. The amplitude of the inductor current for $n = 2$ is:

$$I_2 = \frac{V_2}{Z_2} \approx \frac{V_2}{2\omega L} = \frac{4V_m/3\pi}{2\omega L} = \frac{2V_m}{3\pi\omega L} \qquad (4\text{-}16)$$

For the current to always be positive, the amplitude of the ac term must be less than the dc term (average value). Using the above equations and solving for L,

$$I_2 < I_L$$
$$\frac{2V_m}{3\pi\omega L} < \frac{2V_m}{\pi R}$$
$$L > \frac{R}{3\omega}$$

or

$$\frac{3\omega L}{R} > 1 \qquad \text{for continuous current} \qquad (4\text{-}17)$$

If $3\omega L/R > 1$, the current is continuous and the output voltage is $2V_m/\pi$. Otherwise, the output voltage must be determined from analysis for discontinuous current, discussed as follows.

Discontinuous Current for *LC* Filtered Output For discontinuous inductor current, the current reaches zero during each period of the current waveform (Fig. 4-8*c*). Current becomes positive again when the bridge output voltage reaches the level of the capacitor voltage, which is at $\omega t = \alpha$.

$$\alpha = \sin^{-1}\left(\frac{V_o}{V_m}\right) \qquad (4\text{-}18)$$

While current is positive, the voltage across the inductor is

$$v_L = V_m \sin(\omega t) - V_o \qquad (4\text{-}19)$$

Where the output voltage V_o is yet to be determined. Inductor current is expressed as:

$$i_L(\omega t) = \frac{1}{\omega L}\int_{\alpha}^{\omega t}\left[V_m \sin(\omega t) - V_o\right] d(\omega t)$$

$$= \frac{1}{\omega L}\left[V_m\left(\cos\alpha - \cos\omega t\right)\right] - V_o(\omega t - \alpha) \qquad (4\text{-}20)$$

for $\alpha \le \omega t \le \beta$ when $\beta < \pi$

Which is valid until the current reaches zero, at $\omega t = \beta$.
The solution for the load voltage V_o is based on the fact that the average inductor current must equal the current in the load resistor. Unfortunately, a closed form solution is not available, and an iterative technique is required.

A procedure for determining V_o is as follows:
1. Estimate a value for V_o slightly below V_m and solve for α in Eq. (4-18).
2. Solve for β numerically in Eq. (4-20) for inductor current,

$$i_L(\beta) = 0 = V_m(\cos\alpha - \cos\beta) - V_o(\beta - \alpha)$$

3. Solve for average inductor current I_L.

$$i_L = \frac{1}{\pi}\int_{\alpha}^{\beta} i_L(\omega t)\, d(\omega t)$$

$$= \frac{1}{\pi}\int_{\alpha}^{\beta}\frac{1}{\omega L}\left[V_m(\cos\alpha - \cos\omega t) - V_o(\omega t - \alpha)\right] d(\omega t) \qquad (4\text{-}21)$$

4. Solve for load voltage V_o based upon the average inductor current from step 3.

$$I_R = I_L = \frac{V_o}{R}$$

$V_{o\cdot dis} >> V_{o\cdot con}$

or

$$V_o = I_L R \qquad (4\text{-}22)$$

5. Repeat steps 1 to 4 until the computed value of V_o in step 4 equals the estimated V_o in step 1.
Output voltage for discontinuous current is larger than for continuous current.

If there is no load, the capacitor charges to the peak value of the source so the maximum output is V_m. Figure 4-8d shows normalized output V_o/V_m as a function of $3\omega L/R$.

Example 4-5 (Full-Wave Rectifier with *LC* Filter)
A full-wave rectifier has a source of $v_s(t) = 100\ sin(377t)\ V$. An *LC* filter as in Fig. 4-8*a* is used, with $L = 5\ mH$ and $C = 10{,}000\ \mu F$. The load resistance is (*a*) 5 Ω and (*b*) 50 Ω.
Determine the output voltage for each case.

■ **Solution**

Using Eq. (4-17), continuous inductor current exists when:

$$R < 3\omega L = 3(377)(0.005) = 5.7\ \Omega$$

Which indicates continuous current for 5 Ω and discontinuous current for 50 Ω.

(a) For $R = 5\ \Omega$ with continuous current, output voltage is determined from Eq. (4-14).

$$V_o = \frac{2V_m}{\pi} = \frac{2(100)}{\pi} = 63.7\ \text{V}$$

(b) For $R = 50\ \Omega$ with discontinuous current, the iteration method is used to determine V_o. Initially, V_o is estimated to be 90 V. The results of the iteration are as follows:

Estimated V_o	α	β	Calculated V_o	
90	1.12	2.48	38.8	(Estimate is too high)
80	0.93	2.89	159	(Estimate is too low)
85	1.12	2.70	88.2	(Estimate is slightly low)
86	1.04	2.66	76.6	(Estimate is too high)
85.3	1.02	2.69	84.6	(Approximate solution)

Therefore, V_o is approximately 85.3 V. As a practical matter, three significant figures for the load voltage may not be justified when predicting performance of a real circuit.
Knowing that the output voltage is slightly above 85 V after the third iteration is probably sufficient. Output could also be estimated from the graph of Fig. 4-8d.

4.3 Controlled Full-Wave Rectifiers

A versatile method of controlling the output of a full-wave rectifier is to substitute controlled switches such as thyristors (SCRs) for the diodes. Output is controlled by adjusting the delay angle of each SCR, resulting in an output voltage that is adjustable over a limited range.

Controlled full-wave rectifiers are shown in Fig. 4-10. For the bridge rectifier, SCRs S_1 and S_2 will become forward-biased when the source becomes positive but will not conduct until gate signals are applied. Similarly, S_3 and S_4 will become forward-biased when the source becomes negative but will not conduct until they receive gate signals. For the center-tapped transformer rectifier, S_1 is forward-biased when v_S is positive, and S_2 is forward-biased when *vs* is negative, but each will not conduct until it receives a gate signal.

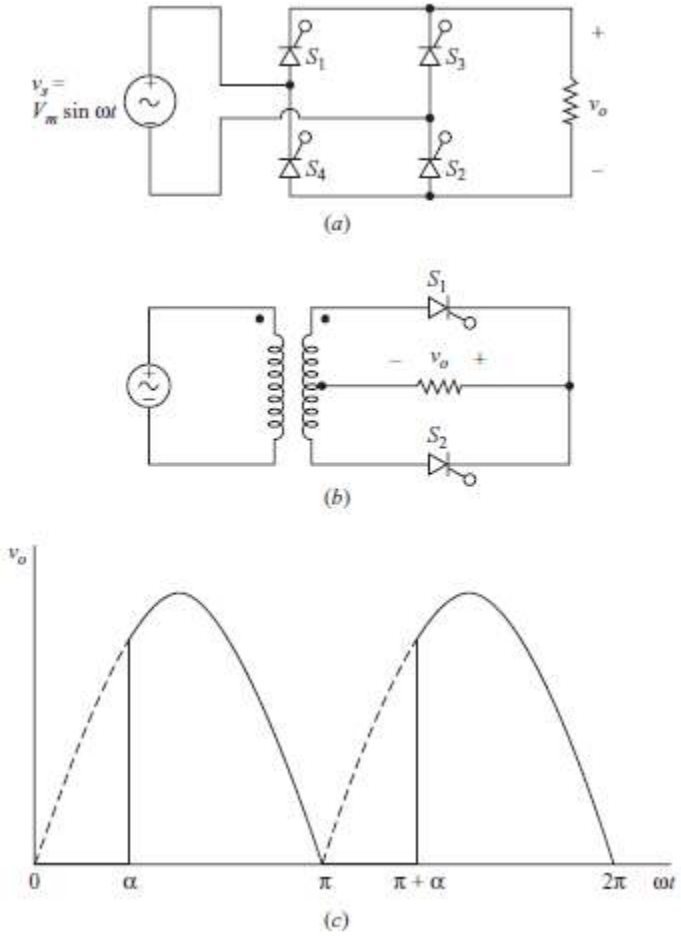


Figure 4-10 (*a*) Controlled full-wave bridge rectifier; (*b*) Controlled full-wave center-tapped transformer rectifier; (*c*) Output for a resistive load.

The delay angle α is the angle interval between the forward biasing of the SCR and the gate signal application. If the delay angle is zero, the rectifiers behave exactly as uncontrolled rectifiers with diodes. The discussion that follows generally applies to both bridge and center-tapped rectifiers.

Resistive Load

The output voltage waveform for a controlled full-wave rectifier with a resistive load is shown in Fig. 4-10*c*. The average component of this waveform is determined from :

$$V_o = \frac{1}{\pi}\int_{\alpha}^{\pi} V_m \sin(\omega t)\, d(\omega t) = \frac{V_m}{\pi}(1 + \cos\alpha) \qquad (4\text{-}23)$$

The average output current is then

$$I_o = \frac{V_o}{R} = \frac{V_m}{\pi R}(1 + \cos\alpha) \qquad (4\text{-}24)$$

The power delivered to the load is a function of the input voltage, the delay angle, and the load components; $P = I_{rms}^2\ R$ is used to determine the power in a resistive load, where:

$$I_{\text{rms}} = \sqrt{\frac{1}{\pi}\int_{\alpha}^{\pi}\left(\frac{V_m}{R}\sin\omega t\right)^2 d(\omega t)}$$

$$= \frac{V_m}{R}\sqrt{\frac{1}{2} - \frac{\alpha}{2\pi} + \frac{\sin(2\alpha)}{4\pi}} \qquad (4\text{-}25)$$

The rms current in the source is the same as the rms current in the load.

Example 4-6(Controlled Full-Wave Rectifier with Resistive Load)
The full-wave controlled bridge rectifier of Fig. 4-10*a* has an AC input of $120\ V\ rms\ at\ 60\ Hz$ and a $20\ \Omega$ load resistor. The delay angle is $40°$. Determine the average current in the load, the power absorbed by the load, and the source volt-amperes, and the power factor.

■ **Solution**

The average output voltage is determined from Eq. (4-23).

$$V_o = \frac{V_m}{\pi}\left(1 + \cos\alpha\right) = \frac{\sqrt{2}\,(120)}{\pi}\left(1 + \cos 40^\circ\right) = 95.4\ \text{V}$$

Average load current is

$$I_o = \frac{V_o}{R} = \frac{95.4}{20} = 4.77\ \text{A}$$

Power absorbed by the load is determined from the *rms* current from Eq. (4-24), remembering to use α in radians.

$$I_{\text{rms}} = \frac{\sqrt{2}(120)}{20}\sqrt{\frac{1}{2} - \frac{0.698}{2\pi} + \frac{\sin[2(0.698)]}{4\pi}} = 5.80\ \text{A}$$

$$P = I_{\text{rms}}^2 R = (5.80)^2\,(20) = 673\ \text{W}$$

The *rms* current in the source is also 5.80 A, and the apparent power of the source is:

$$S = V_{\text{rms}}\, I_{\text{rms}} = (120)(5.80) = 696\ \text{VA}$$

Power factor is

$$\text{pf} = \frac{P}{S} = \frac{672}{696} = 0.967$$

RL Load, Discontinuous Current

The load current for a controlled full-wave rectifier with an *RL* load (Fig. 4-11*a*) can be either continuous or discontinuous, and a separate analysis is required for each.

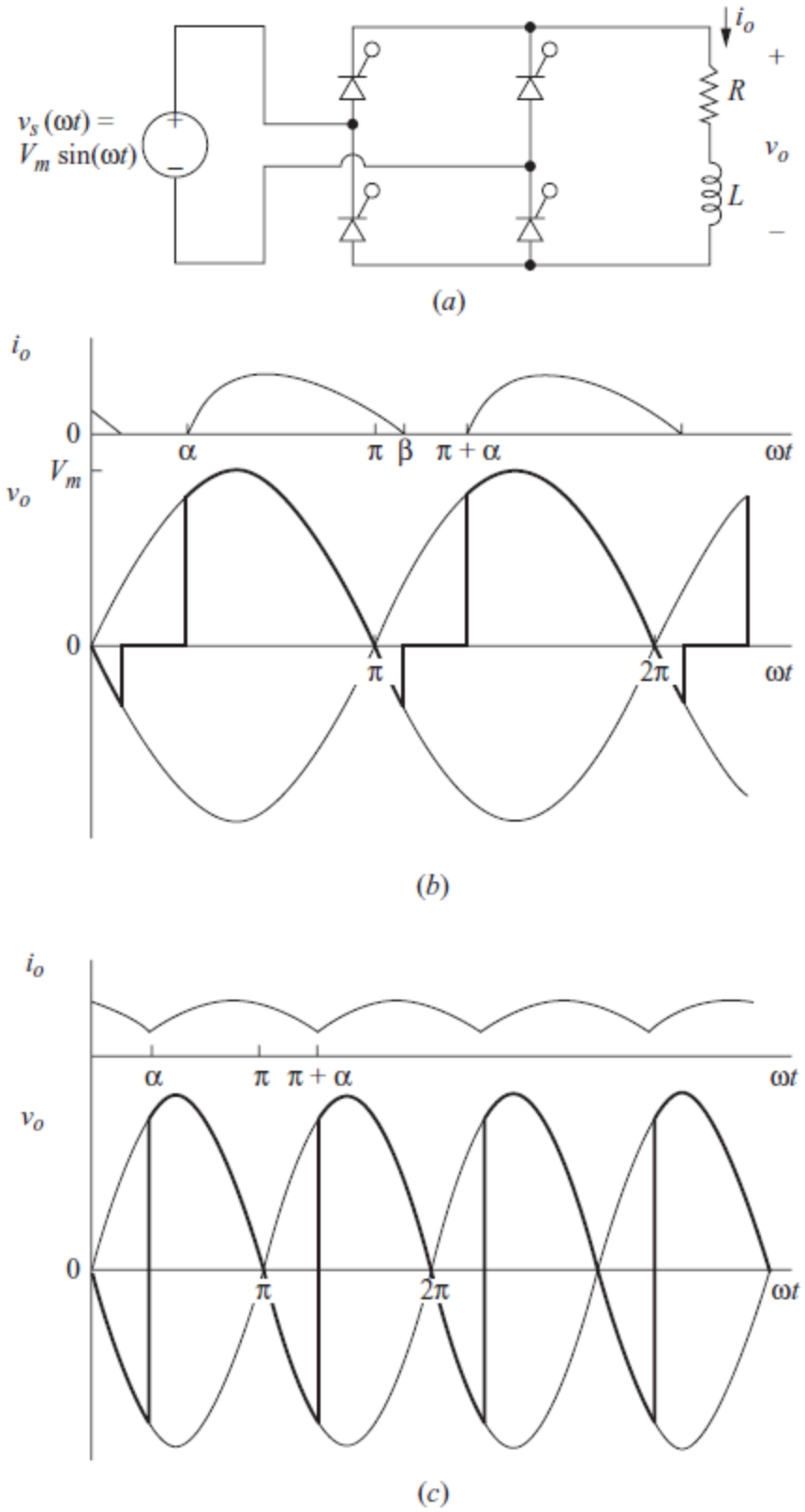


Figure 4-11 (*a*) Controlled rectifier with *RL* load; (*b*) Discontinuous current;

(*c*) Continuous current.

Starting the analysis at $\omega t = 0$ with zero load current, SCRs S_1 and S_2 in the bridge rectifier will be forward-biased and S_3 and S_4 will be reverse-biased as the source voltage becomes positive. Gate signals are applied to S_1 and S_2 at $\omega t = \alpha$, turning S_1 and S_2 on. With S_1 and S_2 on, the load voltage is equal to the source voltage. For this condition, the circuit is identical to that of the controlled half-wave rectifier of Chap. 3, having a current function:

$$i_o(\omega t) = \frac{v_m}{Z}\left[\sin(\omega t - \theta) - \sin(\alpha - \theta)\, e^{-(\omega t - \alpha)/\omega\tau}\right] \qquad \text{for } \alpha \le \omega t \le \beta$$

where (4-26)

$$Z = \sqrt{R^2 + (\omega L)^2} \qquad \theta = \tan^{-1}\left(\frac{\omega L}{R}\right) \qquad \text{and} \qquad \tau = \frac{L}{R}$$

The above current function becomes zero at $\omega t = \beta$. If $\beta < \pi + \alpha$, the current remains at zero until $\omega t = \pi + \alpha$ when gate signals are applied to S_3 and S_4 which are then forward-biased and begin to conduct. This mode of operation is called *discontinuous current*, which is illustrated in Fig. 4-11*b*.

$$\beta < \alpha + \pi \rightarrow \text{discontinuous current} \qquad (4\text{-}27)$$

Analysis of the controlled full-wave rectifier operating in the discontinuous current mode is identical to that of the controlled half-wave rectifier except that the period for the output current is π rather than 2π rad.

Example 4-7(Controlled Full-Wave Rectifier, Discontinuous Current)
A controlled full-wave bridge rectifier of Fig. 4-11a has a source of $120\ V\ rms\ at\ 60\ Hz$, $R = 10\ \Omega, L = 20\ mH$, and $\alpha = 60°$. Determine (a) an expression for load current, (b) the average load current, and (c) the power absorbed by the load.

■ Solution

From the parameters given,

$$V_m = \sqrt{2} \times 120 = 169.7\ V$$

$$Z = \sqrt{R^2 + (\omega L)^2} = \sqrt{10^2 + [(377)(0.02)]^2} = 12.5\ \Omega$$

$$\theta = \tan^{-1}\left(\frac{\omega L}{R}\right) = \tan^{-1}\left[\frac{(377)(0.02)}{10}\right] = 0.646 \text{ rad}$$

$$\omega\tau = \frac{\omega L}{R} = \frac{(377)(0.02)}{10} = 0.754 \text{ rad}$$

$$\alpha = 60° = 1.047 \text{ rad}$$

(a) Substituting into Eq. (4-26),

$$i_o(\omega t) = 13.6 \sin(\omega t - 0.646) - 21.2e^{-\omega t/0.754} \text{ A} \qquad \text{for } \alpha \le \omega t \le \beta$$

Solving $i_o(\beta) = 0$ numerically for β, $\beta = 3.78\, rad = (216°)$. Since $\pi + \beta = 3.14 + 1.047 = 4.19 > \beta$, the current is discontinuous, and the above expression for current is valid.

(b) Average load current is determined from the numerical integration of:

$$I_o = \frac{1}{\pi}\int_{\alpha}^{\beta} i_o(\omega t)\, d(\omega t) = 7.05 \text{ A}$$

(c) Power absorbed by the load occurs in the resistor and is computed from $I_{rms}^2\ R$, where:

$$I_{\text{rms}} = \sqrt{\frac{1}{\pi}\int_{\alpha}^{\beta} i_o(\omega t)\, d(\omega t)} = 8.35 \text{ A}$$

$$P = (8.35)^2(10) = 697 \text{ W}$$

RL Load, Continuous Current

If the load current is still positive at $\omega t = \pi + \alpha$when gate signals are applied to S_3 and S_4 in the above analysis, S_3 and S_4 are turned on and $S_1$and S_2 are forced off. Since the initial condition for current in the second half-cycle is not zero, the current function does not repeat. Equation (4-26) is not valid in the steady state for continuous current. For an *RL* load with continuous current, the steady-state current and voltage waveforms are generally as shown in Fig. 4-11*c*.

The boundary between continuous and discontinuous current occurs when β for Eq. (4-26) is $\pi + \alpha$. The current at $\omega t = \pi + \alpha$ must be greater than zero for continuous-current operation.

$$i(\pi + \alpha) \geq 0$$

$$\sin(\pi + \alpha - \theta) - \sin(\pi + \alpha - \theta)\, e^{-(\pi+\alpha-\alpha)/\omega\tau} \geq 0$$

Using

$$\sin(\pi + \alpha - \theta) = \sin(\theta - \alpha)$$

$$\sin(\theta - \alpha)\left(1 - e^{-(\pi/\omega\tau)}\right) \geq 0$$

Solving for α,

$$\alpha \leq \theta$$

Using

$$\theta = \tan^{-1}\left(\frac{\omega L}{R}\right)$$

$$\alpha \leq \tan^{-1}\left(\frac{\omega L}{R}\right) \qquad \text{for continuous current} \qquad (4\text{-}28)$$

Either Eq. (4-27) or Eq. (4-28) can be used to check whether the load current is continuous or discontinuous.

Amethod for determining the output voltage and current for the continuous current case is to use the Fourier series. The Fourier series for the voltage waveform for the continuous-current case shown in Fig. 4-11*c* is expressed in general form as:

$$v_o(\omega t) = V_o + \sum_{n=1}^{\infty} V_n \cos(n\omega_0 t + \theta_n) \qquad (4\text{-}29)$$

The DC (average) value is

$$V_o = \frac{1}{\pi}\int_{\alpha}^{\alpha+\pi} V_m \sin(\omega t)\, d(\omega t) = \frac{2V_m}{\pi}\cos\alpha \qquad (4\text{-}30)$$

The amplitudes of the AC terms are calculated from

$$V_n = \sqrt{a_n^2 + b_n^2} \tag{4-31}$$

Where

$$a_n = \frac{2V_m}{\pi}\left[\frac{\cos(n+1)\alpha}{n+1} - \frac{\cos(n-1)\alpha}{n-1}\right]$$

$$b_n = \frac{2V_m}{\pi}\left[\frac{\sin(n+1)\alpha}{n+1} - \frac{\sin(n-1)\alpha}{n-1}\right] \tag{4-32}$$

$$n = 2, 4, 6, \ldots$$

Figure 4-12 shows the relationship between normalized harmonic content of the output voltage and delay angle.

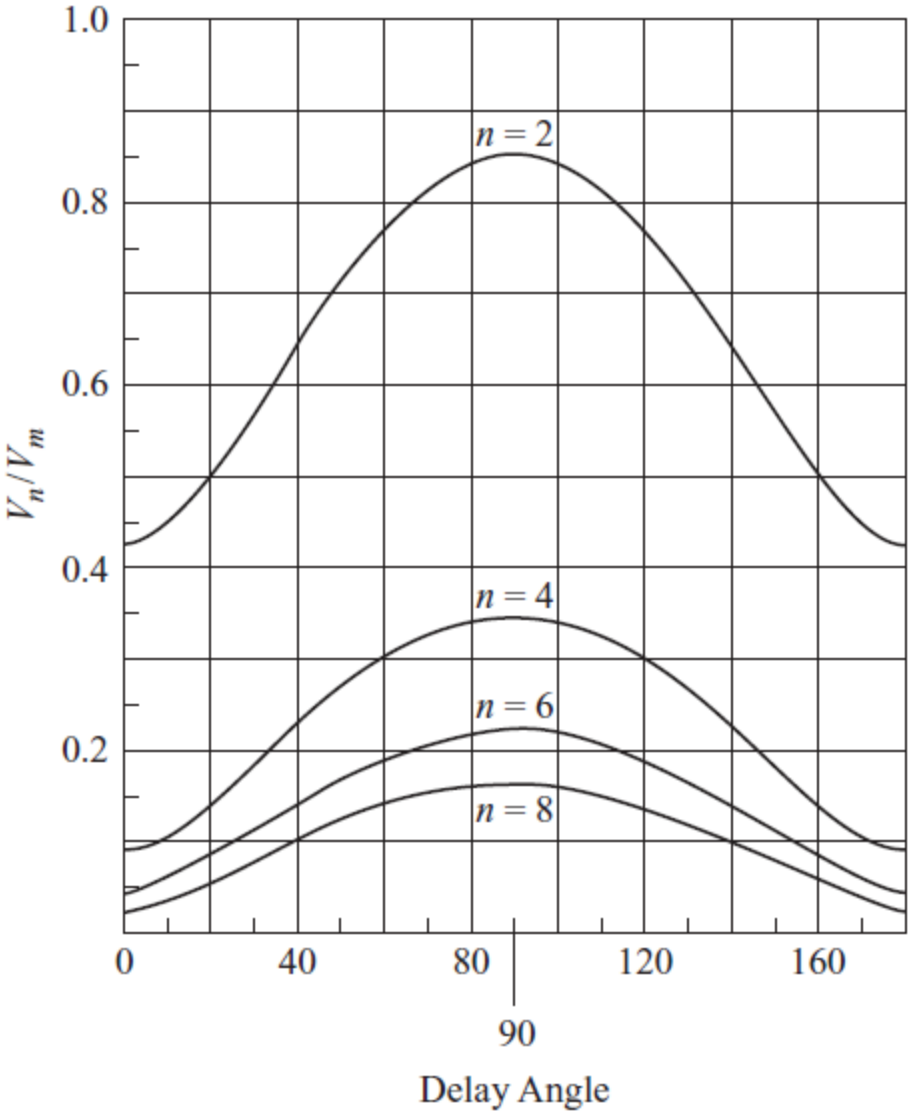


Figure 4-12 Output harmonic voltages as a function of delay angle for a single-phase controlled rectifier.

The Fourier series for current is determined by superposition as was done for the uncontrolled rectifier earlier in this chapter. The current amplitude at each frequency is determined from Eq. (4-5). The rms current is determined by combining the rms currents at each frequency. From Eq. (2-43),

$$I_{\text{rms}} = \sqrt{I_o^2 + \sum_{n=2,4,6\ldots}^{\infty} \left(\frac{I_n}{\sqrt{2}}\right)^2}$$

Where

$$I_o = \frac{V_o}{R} \qquad \text{and} \qquad I_n = \frac{V_n}{Z_n} = \frac{V_n}{|R + jn\omega_0 L|} \tag{4-33}$$

As the harmonic number increases, the impedance for the inductance increases.

Therefore, it may be necessary to solve for only a few terms of the series to be able to calculate the rms current. If the inductor is large, the ac terms will become small, and the current will essentially be DC.

Example 4-8(Controlled Full-Wave Rectifier with *RL* Load, Continuous Current)

A controlled full-wave bridge rectifier of Fig. 4-11a has a source of $120\ V\ rms\ at\ 60\ Hz$, an RL load where $R = 10\ \Omega$ and $L = 100\ mH$. The delay angle $\alpha = 60°$ (same as Example 4-7 except L is larger). (a) Verify that the load current is continuous. (b) Determine the DC (average) component of the current. (c) Determine the power absorbed by the load.

■ Solution

(a) Equation (4-28) is used to verify that the current is continuous.

$$\tan^{-1}\left(\frac{\omega L}{R}\right) = \tan^{-1}\left[\frac{(377)(0.1)}{10}\right] = 75°$$

$$\alpha = 60° < 75° \qquad \therefore \text{ continuous current}$$

(b) The voltage across the load is expressed in terms of the Fourier series of Eq. (4-29).

The dc term is computed from Eq. (4-30).

$$V_0 = \frac{2V_m}{\pi}\cos\alpha = \frac{2\sqrt{2}(120)}{\pi}\cos(60°) = 54.0\ \text{V}$$

(c) The amplitudes of the ac terms are computed from Eqs. (4-31) and (4-32) and are summarized in the following table where, $Z_n = |R + j\omega L|\ and\ I_n = V_n/Z_n$.

n	a_n	b_n	V_n	Z_n	I_n
0 (dc)	—	—	54.0	10	5.40
2	−90	−93.5	129.8	76.0	1.71
4	46.8	−18.7	50.4	151.1	0.33
6	−3.19	32.0	32.2	226.4	0.14

The rms current is computed from Eq. (4-33).

$$I_{\text{rms}} = \sqrt{(5.40)^2 + \left(\frac{1.71}{\sqrt{2}}\right)^2 + \left(\frac{0.33}{\sqrt{2}}\right)^2 + \left(\frac{0.14}{\sqrt{2}}\right)^2 + \cdots} \approx 5.54 \text{ A}$$

Power is computed from $I_{rms}^2 R$.

$$P = (5.54)^2(10) = 307 \text{ W}$$

Note that the rms current could be approximated accurately from the dc term and one AC term ($n = 2$). Higher-frequency terms are very small and contribute little to the power in the load.

Controlled Rectifier with *RL*-Source Load

The controlled rectifier with a load that is a series resistance, inductance, and DC voltage (Fig. 4-14) is analyzed much like the uncontrolled rectifier of Fig. 4-5*a* discussed earlier in this chapter.

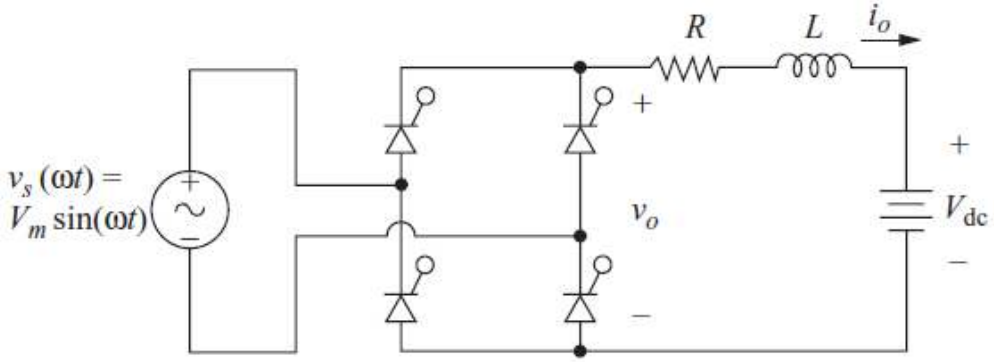


Figure 4-14 Controlled rectifier with *RL*-source load.

For the controlled rectifier, the SCRs may be turned on at any time that they are forward-biased, which is at an angle

$$\alpha \geq \sin^{-1}\left(\frac{V_{\text{dc}}}{V_m}\right) \qquad (4\text{-}34)$$

For the continuous-current case, the bridge output voltage is the same as in Fig. 4-11*c*. The average bridge output voltage is:

$$V_o = \frac{2V_m}{\pi} \cos \alpha \qquad (4\text{-}35)$$

The average load current is

$$I_o = \frac{V_o - V_{\text{dc}}}{R} \qquad (4\text{-}36)$$

The ac voltage terms are unchanged from the controlled rectifier with an RL load in Fig. 4-11a and are described by Eqs. (4-29) to (4-32). The ac current terms are determined from the circuit of Fig. 4-14c. Power absorbed by the DC voltage is:

$$P_{\text{dc}} = I_o V_{\text{dc}} \qquad (4\text{-}37)$$

Power absorbed by the resistor in the load is $I_{rms}^2 \ R$. If the inductance is large and the load current has little ripple, power absorbed by the resistor is approximately $I_o^2 \ R$.

Example 4-10(Controlled Rectifier with *RL*-Source Load)
The controlled rectifier of Fig. 4-14 has an ac source of $240\ V\ rms\ at\ 60\ Hz$, $V_{dc} = 100\ V$, $R = 5\ \Omega$, and an inductor large enough to cause continuous current. (*a*) Determine the delay angle α such that the power absorbed by the dc source is 1000 W. (*b*) Determine the value of inductance that will limit the peak-to-peak load current variation to 2 A.

■ **Solution**

(*a*) For the power in the 100-V dc source to be 1000 W, the current in it must be 10 A.
The required output voltage is determined from Eq. (4-36) as:

$$V_o = V_{dc} + I_o R = 100 + (10)(5) = 150 \ \text{V}$$

The delay angle which will produce a 150 V dc output from the rectifier is determined from Eq. (4-35).

$$\alpha = \cos^{-1}\left(\frac{V_o \pi}{2V_m}\right) = \cos^{-1}\left[\frac{(150)(\pi)}{2\sqrt{2}(240)}\right] = 46°$$

(b) Variation in load current is due to the ac terms in the Fourier series. The load current amplitude for each of the ac terms is

$$I_n = \frac{V_n}{Z_n}$$

where V_n is described by Eqs. (4-31) and (4-32) or can be estimated from the graph of Fig. 4-12. The impedance for the ac terms is:

$$Z_n = |R + jn\omega_0 L|$$

Since the decreasing amplitude of the voltage terms and the increasing magnitude of the impedance both contribute to diminishing ac currents as n increases, the peak-to-peak current variation will be estimated from the first AC term. For $n = 2$, V_n/V_m is estimated from Fig. 4-12 as 0.68 for $\alpha = 46°$, making $V_2 = 0.68\, V_m = 0.68\ (240\ )\sqrt{2} = 230\ V$.

The peak-to-peak variation of 2 A corresponds to a 1A zero-to-peak amplitude. The required load impedance for $n = 2$ is then:

$$Z_2 = \frac{V_2}{I_2} = \frac{230\ \text{V}}{1\ \text{A}} = 230\ \Omega$$

The 5 Ω resistor is insignificant compared to the total 230 Ω required impedance, so $Z_n \approx n\omega L$. Solving for L,

$$L \approx \frac{Z_2}{2\omega} = \frac{230}{2(377)} = 0.31\ \text{H} \qquad L > 0,31$$

A slightly larger inductance should be chosen to allow for the effect of higher-order AC terms.

Controlled Single-Phase Converter Operating as an Inverter

The above discussion focused on circuits operating as rectifiers, which means that the power flow is from the ac source to the load. It is also possible for power to flow from the load to the ac source, which classifies the circuit as an inverter.

For inverter operation of the converter in Fig. 4-14, power is supplied by the DC source, and power is absorbed by the bridge and is transferred to the ac system.

The load current must be in the direction shown because of the SCRs in the bridge. For power to be supplied by the DC source, V_{dc} must be negative. For power to be absorbed by the bridge and transferred to the AC system, the bridge output voltage V_o must also be negative. Equation (4-35) applies, so a delay angle larger than 90° will result in a negative output voltage.

$$\begin{aligned} 0 < \alpha < 90° &\rightarrow V_o > 0 \quad \text{rectifier operation} \\ 90° < \alpha < 180° &\rightarrow V_o < 0 \quad \text{inverter operation} \end{aligned} \tag{4-38}$$

The voltage waveform for $\alpha = 150°$ and continuous inductor current is shown in Fig. 4-15. Equations (4-36) to (4-38) apply. If the inductor is large enough to effectively eliminate the AC current terms and the bridge is

عوض بـ V_{dc} وبالبقية دغيها

lossless, the power absorbed by the bridge and transferred to the AC system is

$$P_{\text{bridge}} = P_{\text{ac}} = -I_o V_o \qquad (4\text{-}39)$$

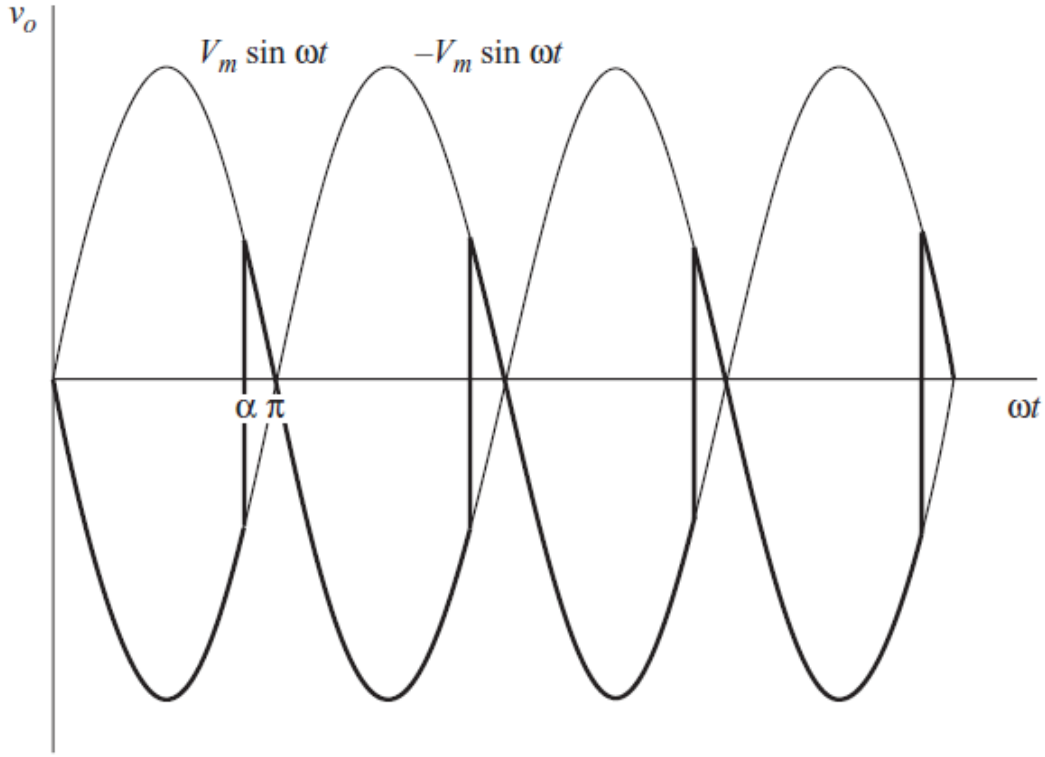


Figure 4-15 Output voltage for the controlled single-phase converter of Fig. 4-14 operating as an inverter, $\alpha = 150°$ and $V_{dc} < 0$.

كمان بتجي $+ -V_{dc}$

Example 4-11(Single-Phase Bridge Operating as an Inverter)
The dc voltage in Fig. 4-14 represents the voltage generated by an array of solar cells and has a value of 110 V, connected such that $V_{dc} = 110\ V$. The solar cells are capable of producing 1000 W. The AC source is 120 $V\ rms$, $R = 0.5\Omega$, and L is large enough to cause the load current to be essentially DC. Determine the delay angle α such that 1000 W is supplied by the solar cell array. Determine the power transferred to the AC system and the losses in the resistance. Assume ideal SCRs.

■ **Solution**
For the solar cell array to supply 1000 W, the average current must be:

بتشوف نوع الدارة وبترجع لقوانينها

$$I_o = \frac{P_{dc}}{V_{dc}} = \frac{1000}{110} = 9.09 \text{ A}$$

The average output voltage of the bridge is determined from Eq. (4-36).

$$V_o = I_o R + V_{dc} = (9.09)(0.5) + (-110) = -105.5 \text{ V}$$

The required delay angle is determined from Eq. (4-35).

$$\alpha = \cos^{-1}\left(\frac{V_o \pi}{2V_m}\right) = \cos^{-1}\left[\frac{-105.5\pi}{2\sqrt{2}(120)}\right] = 165.5°$$

Power absorbed by the bridge and transferred to the ac system is determined from
Eq. (4-39).

$$P_{ac} = -V_o I_o = (-9.09)(-105.5) = 959 \text{ W}$$

Power absorbed by the resistor is

$$P_R = I_{rms}^2 R \approx I_o^2 R = (9.09)^2(0.5) = 41 \text{ W}$$

Note that the load current and power will be sensitive to the delay angle and the voltage drops across the SCRs because bridge output voltage is close to the dc source voltage. For example, assume that the voltage across a conducting SCR is 1 V. Two SCRs conduct at all times, so the average bridge output voltage is reduced to

$$V_o = -105.5 - 2 = -107.5 \text{ V}$$

2 SCR

Average load current is then

$$I_o = \frac{-107.5 - (-110)}{0.5} = 5.0 \text{ A}$$

Power delivered to the bridge is then reduced to

$$P_{bridge} = (107.5)(5.0) = 537.5 \text{ W}$$

Average current in each SCR is one-half the average load current. Power absorbed by each SCR is approximately

$$P_{SCR} = I_{SCR} V_{SCR} = \frac{1}{2} I_o V_{SCR} = \frac{1}{2}(5)(1) = 2.5 \text{ W}$$

Total power loss in the bridge is then $4(2.5) = 10\ W$, and power delivered to the ac source is $537.5 - 10 = 527.5\ W$.

4.4 THREE-PHASE RECTIFIERS

Three-phase rectifiers are commonly used in industry to produce a dc voltage and current for large loads. The three-phase full-bridge rectifier is shown in Fig. 4-16*a*.

The three-phase voltage source is balanced and has phase sequence *a-b-c*. The source and the diodes are assumed to be ideal in the initial analysis of the circuit.

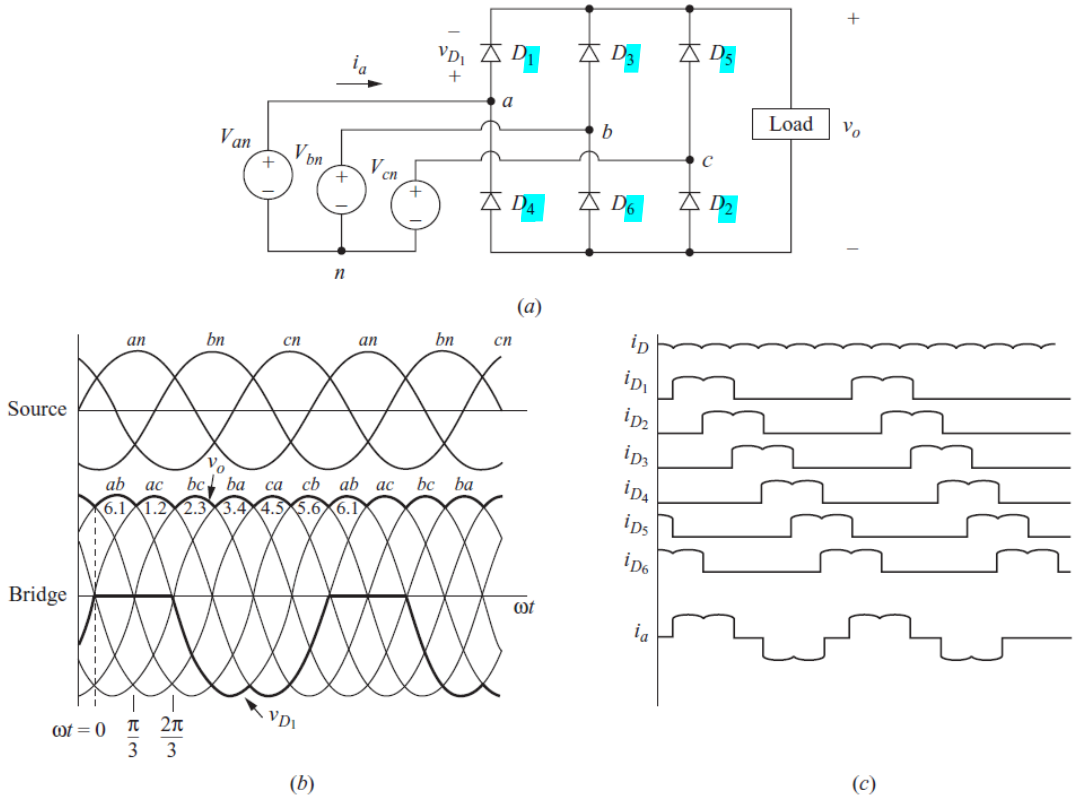


Figure 4-16 (*a*) Three-phase full-bridge rectifier; (*b*) Source and output voltages; (*c*) Currents for a resistive load.

Some basic observations about the circuit are as follows:

1. Kirchhoff's voltage law around any path shows that only one diode in the top half of the bridge may conduct at one time ($D_1, D_3, or\ D_5$). The diode that is conducting will have its anode connected to the phase voltage that is highest at that instant.

2. Kirchhoff's voltage law also shows that only one diode in the bottom half of the bridge may conduct at one time ($D_2, D_4, or\ D_6$).. The diode that is conducting will have its cathode connected to the phase voltage that is lowest at that instant.

3. As a consequence of items 1 and 2 above, D_1 and D_4 cannot conduct at the same time. Similarly, D_3 and D_6 cannot conduct simultaneously, nor can D_5 and D_2.

4. The output voltage across the load is one of the line-to-line voltages of the source. For example, when D_1 and D_2 are on, the output voltage is v_{ac}. Furthermore, the diodes that are on are determined by which line-to-line voltage is the highest at that instant. For example, when v_{ac} is the highest line-to-line voltage, the output is v_{ac}.

5. There are six combinations of line-to-line voltages (three phases taken two at a time). Considering one period of the source to be 360°, a transition of the highest line-to-line voltage must take place every $360°/6 = 60°$. Because of the six transitions that occur for each period of the source voltage, the circuit is called a *six-pulse rectifier*.

6. The fundamental frequency of the output voltage is 6ω, where ω is the frequency of the three-phase source.

Figure 4-16*b* shows the phase voltages and the resulting combinations of line-to-line voltages from a balanced three-phase source. The current in each of the bridge diodes for a resistive load is shown in Fig. 4-16*c*. The diodes conduct in pairs (6,1), (1,2), (2,3), (3,4), (4,5), (5,6), (6,1), ... Diodes turn on in the sequence 1, 2, 3, 4, 5, 6, 1, ...

The current in a conducting diode is the same as the load current. To determine the current in each phase of the source, Kirchhoff's current law is applied at nodes *a, b,* and *c,*

$$\begin{aligned} i_a &= i_{D_1} - i_{D_4} \\ i_b &= i_{D_3} - i_{D_6} \\ i_c &= i_{D_5} - i_{D_2} \end{aligned} \tag{4-40}$$

Since each diode conducts one-third of the time, resulting in:

$$\boxed{\begin{aligned} I_{D,\text{avg}} &= \frac{1}{3} I_{o,\text{avg}} \\ I_{D,\text{rms}} &= \frac{1}{\sqrt{3}} I_{o,\text{rms}} \\ I_{s,\text{rms}} &= \sqrt{\frac{2}{3}}\, I_{o,\text{rms}} \end{aligned}} \tag{4-41}$$

The apparent power from the three-phase source is

$$S = \sqrt{3}\, V_{L-L,\text{rms}}\, I_{S,\text{rms}} \tag{4-42}$$

The maximum reverse voltage across a diode is the peak line-to-line voltage. The voltage waveform across diode D_1 is shown in Fig. 4-16b. When D_1 conducts, the voltage across it is zero. When D_1 is off, the output voltage is v_{ab} when D_3 is on and is v_{ac} whenD_5 is on.

The periodic output voltage is defined as $v_o(\omega t) = V_{m,L_L}\ sin(\omega t)$ for $\frac{\pi}{3} \leq \omega t \leq \frac{2\pi}{3}$ with period $\pi/3$ for the purpose of determining the Fourier series coefficients. The coefficients for the sine terms are zero from symmetry, enabling the Fourier series for the output voltage to be expressed as:

$$v_o(t) = V_o + \sum_{n=6,12,18\ldots}^{\infty} V_n \cos(n\omega_0 t + \pi) \qquad (4\text{-}43)$$

The average or DC value of the output voltage is

$$V_0 = \frac{1}{\pi/3}\int_{\pi/3}^{2\pi/3} V_{m,L-L}\sin(\omega t)\,\mathrm{d}(\omega t) = \frac{3V_{m,L-L}}{\pi} = 0.955\,V_{m,L-L} \qquad (4\text{-}44)$$

Where V_{m,L_L} is the peak line-to-line voltage of the three-phase source, which is $V_{,L-L\,,rms}$. The amplitudes of the ac voltage terms are:

$$V_n = \frac{6V_{m,L-L}}{\pi(n^2-1)} \qquad n = 6,\ 12,\ 18,\ \ldots \qquad (4\text{-}45)$$

Since the output voltage is periodic with period one-sixth of the ac supply voltage, the harmonics in the output are of order $6k_$, $k = 1, 2, 3$ .. . An advantage of the three-phase rectifier over the single-phase rectifier is that the output is inherently like a dc voltage, and the high-frequency low-amplitude harmonics enable filters to be effective.

In many applications, a load with series inductance results in a load current that is essentially DC. For a DC load current, the diode and AC line currents are shown in Fig. 4-17. The Fourier series of the currents in phase *a* of the AC line is:

$$i_a(t) = \frac{2\sqrt{3}}{\pi} I_o\left(\cos\omega_0 t - \frac{1}{5}\cos 5\omega_0 t + \frac{1}{7}\cos 7\omega_0 t - \frac{1}{11}\cos 11\omega_0 t + \frac{1}{13}\cos 13\omega_0 t - \cdots\right) \qquad (4\text{-}46)$$

Which consists of terms at the fundamental frequency of the ac system and harmonics of order $6k \pm 1, k = 1, 2, 3, \ldots$.

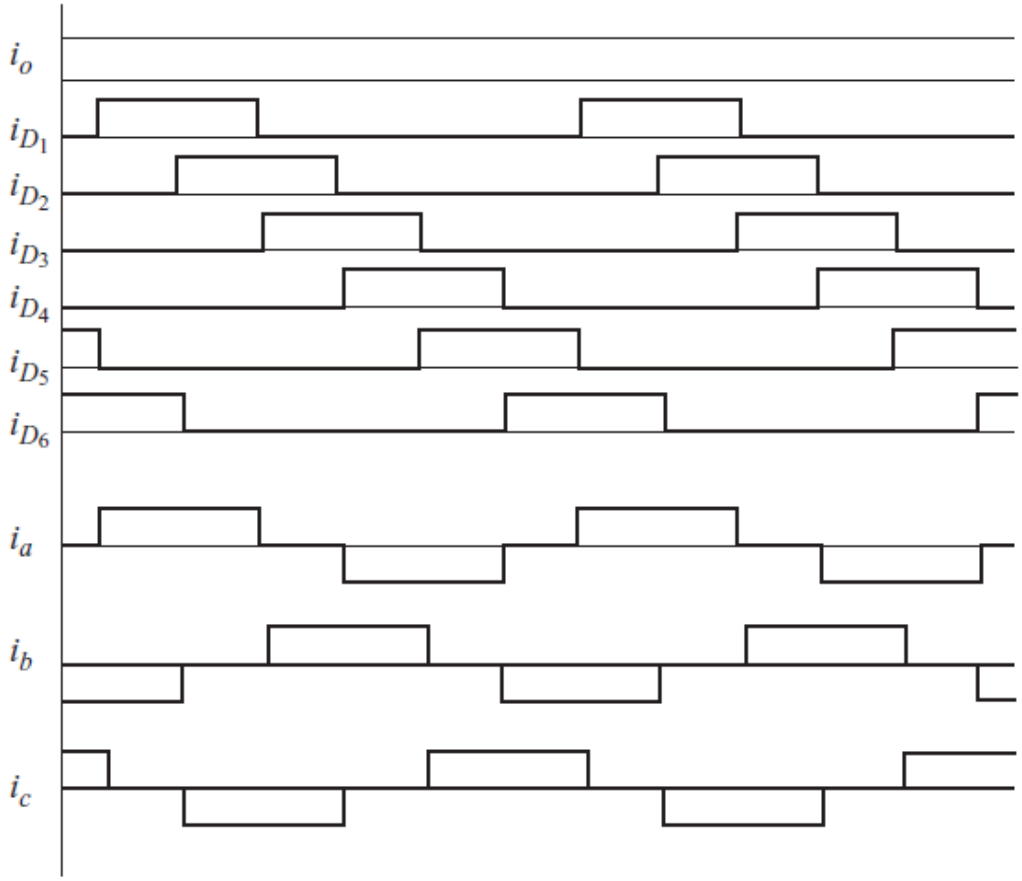


Figure 4-17 Three-phase rectifier currents when the output is filtered.

Because these harmonic currents may present problems in the ac system, filters are frequently necessary to prevent these harmonics from entering the ac system.

Atypical filtering scheme is shown in Fig. 4-18. Resonant filters are used to provide a path to ground for the fifth and seventh harmonics, which are the two lowest and are the strongest in amplitude. Higher-order harmonics are reduced with the high-pass filter. These filters prevent the harmonic currents from propagating through the ac power system. Filter components are chosen such that the impedance to the power system frequency is large.

Example 4-12 (Three-Phase Rectifier)

The three-phase rectifier of Fig. 4-16a has a three-phase source of $480\ V\ rms$ line-to-line, and the load is a $25\ \Omega$ resistance in series with a $50\ mH$ inductance. Determine (a) the DC level of the output voltage, (b) the DC and first AC term of the load current, (c) the average and rms current in the diodes, (d) the rms current in the source, and (e) the apparent power from the source.

■ **Solution**

(*a*) The DC output voltage of the bridge is obtained from Eq. (4-44).

$$V_o = \frac{3V_{m,L-L}}{\pi} = \frac{3\sqrt{2}\,(480)}{\pi} = 648 \text{ V}$$

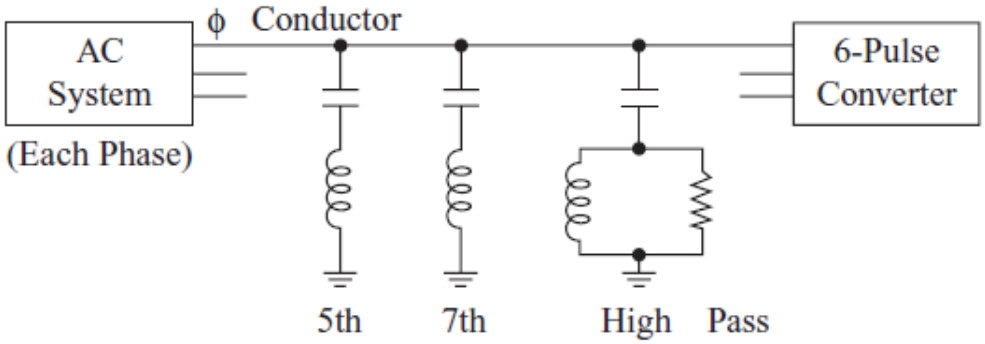


Figure 4-18 Filters for ac line harmonics.

(b)The average load current is

$$I_o = \frac{V_o}{R} = \frac{648}{25} = 25.9 \text{ A}$$

The first ac voltage term is obtained from Eq. (4-45) with $n = 6$, and current is:

$$I_6 = \frac{V_6}{Z_6} = \frac{0.0546V_m}{\sqrt{R^2 + (6\omega L)^2}} = \frac{0.0546\sqrt{2}(480)}{\sqrt{25^2 + [6(377)(0.05)]^2}} = \frac{37.0 \text{ V}}{115.8\ \Omega} = 0.32 \text{ A}$$

$$I_{6,\text{rms}} = \frac{0.32}{\sqrt{2}} = 0.23 \text{ A}$$

This and other ac terms are much smaller than the dc term and can be neglected.

(c)Average and *rms* diode currents are obtained from Eq. (4-41). The *rms* load current is approximately the same as average current since the AC terms are small.

$$I_{D,\text{avg}} = \frac{I_o}{3} = \frac{25.9}{3} = 8.63 \text{ A}$$

$$I_{D,\text{rms}} = \frac{I_{o,\text{rms}}}{\sqrt{3}} \approx \frac{25.9}{\sqrt{3}} = 15.0 \text{ A}$$

(d) The *rms* source current is also obtained from Eq. (4-41).

$$I_{s,\text{rms}} = \left(\sqrt{\frac{2}{3}}\right) I_{o,\text{rms}} \approx \left(\sqrt{\frac{2}{3}}\right) 25.9 = 21.2 \text{ A}$$

(e) The apparent power from the source is determined from Eq. (4-42).

$$S = \sqrt{3}(V_{L-L,\text{rms}})(I_{s,\text{rms}}) = \sqrt{3}\,(480)(21.2) = 17.6 \text{ kVA}$$

4.5 Controlled Three-Phase Rectifiers

The output of the three-phase rectifier can be controlled by substituting SCRs for diodes. Figure 4-20*a* shows a controlled six-pulse three-phase rectifier. With SCRs, conduction does not begin until a gate signal is applied while the SCR is forward-biased. Thus, the transition of the output voltage to the maximum instantaneous line-to-line source voltage can be delayed. The delay angle _ is referenced from where the SCR would begin to conduct if it were a diode. The delay angle is the interval between when the SCR becomes forward-biased and when the gate signal is applied. Figure 4-20*b* shows the output of the controlled rectifier for a delay angle of 45°.

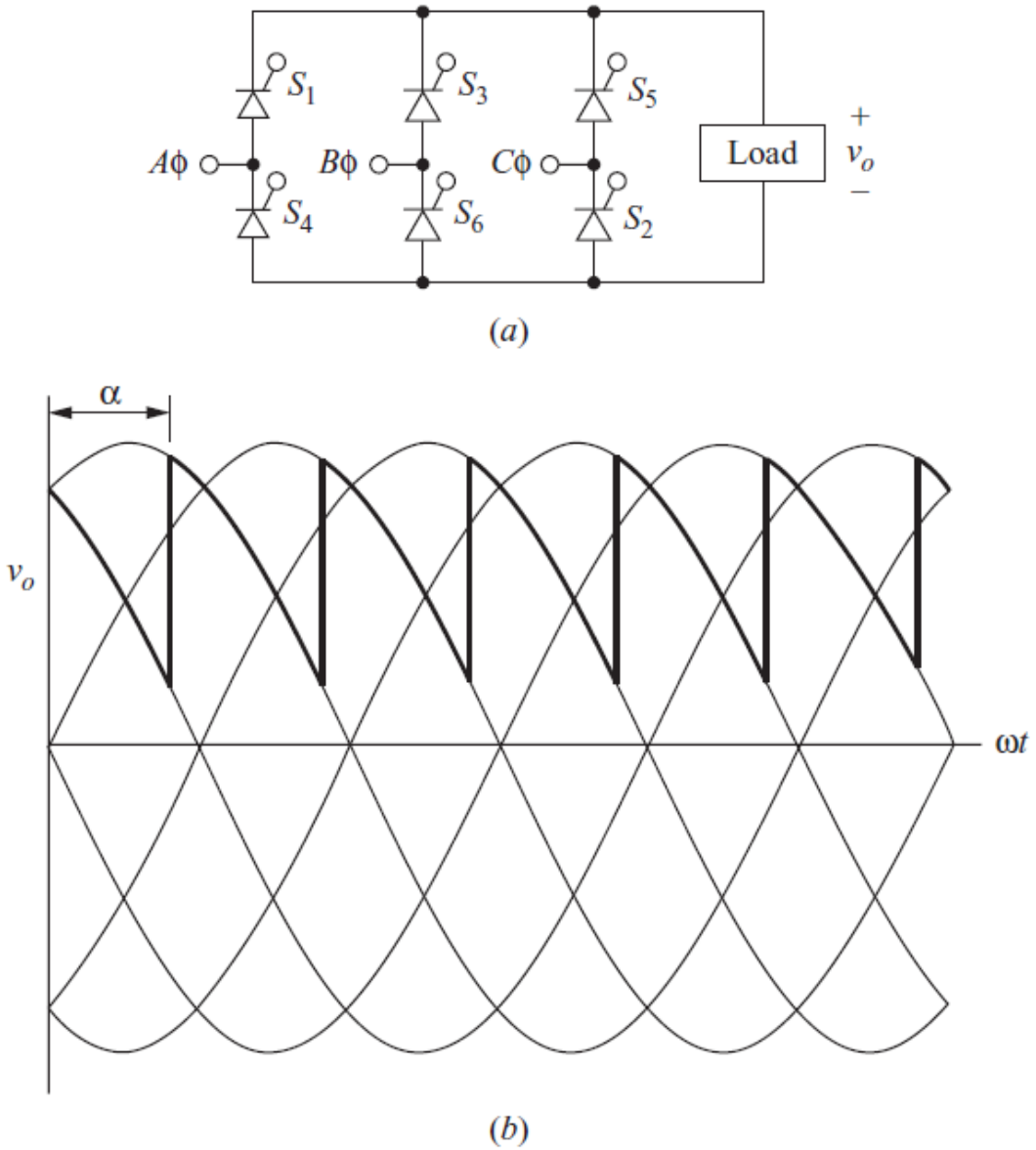


Figure 4-20 (*a*) A controlled three-phase rectifier; (*b*) Output voltage for $\alpha = 45°$.

The average output voltage is:

$$V_o = \frac{1}{\pi/3} \int_{\pi/3+\alpha}^{2\pi/3+\alpha} V_{m,L-L} \sin(\omega t)\, d(\omega t) = \frac{3V_{m,L-L}}{\pi} \cos\alpha \qquad (4\text{-}47)$$

Equation (4-47) shows that the average output voltage is reduced as the delay angle α increases.

Harmonics for the output voltage remain of order $6k$, but the amplitudes are functions of α. Figure 4-21 shows the first three normalized harmonic amplitudes.

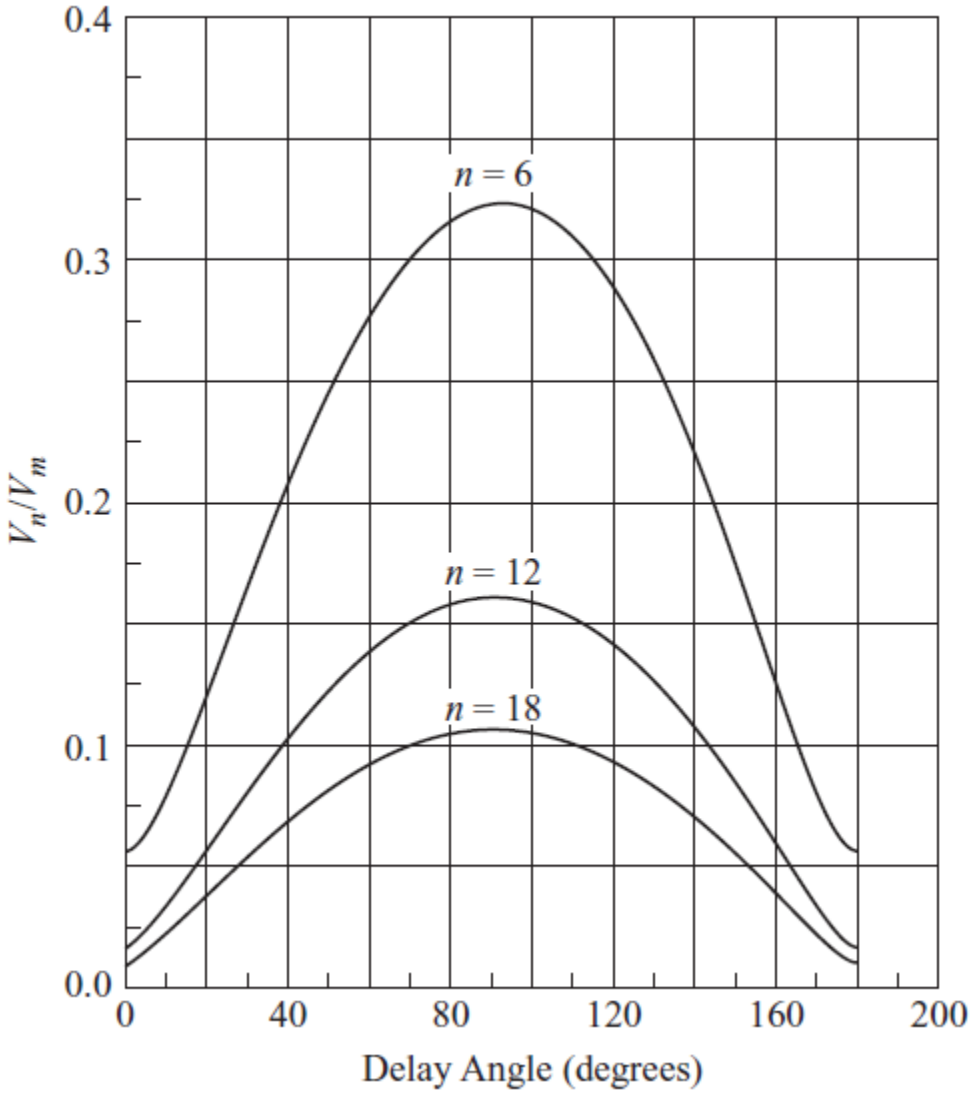


Figure 4-21 Normalized output voltage harmonics as a function of delay angle for a three-phase rectifier.

Example 4-13 (A Controlled Three-Phase Rectifier)

Athree-phase controlled rectifier has an input voltage which is $480\ V\ rms$ at 60 Hz. The load is modeled as a series resistance and inductance with $R = 10\ \Omega\ and\ L = 50\ mH$.

(a) Determine the delay angle required to produce an average current of 50 A in the load. (b) Determine the amplitude of harmonics $n = 6$ and $n = 12$.

■ **Solution**

(a) The required dc component in the bridge output voltage is

$$V_o = I_o R = (50)(10) = 500 \text{ V}$$

Equation (4-47) is used to determine the required delay angle:

$$\alpha = \cos^{-1}\left(\frac{V_o \pi}{3V_{m,L-L}}\right) = \cos^{-1}\left(\frac{500\pi}{3\sqrt{2}(480)}\right) = 39.5°$$

(b) Amplitudes of harmonic voltages are estimated from the graph in Fig. 4-21. For $\alpha = 39.5°$, normalized harmonic voltages are $V_6 / V_m \approx 0.21$ and $V_{12} / V_m \approx 0.10$.

Using $V_m = \sqrt{2}(480)$, $V_6 = 143\ V$, and $V_{12} = 68\ V$, harmonic currents are then:

$$I_6 = \frac{V_6}{Z_6} = \frac{143}{\sqrt{10^2 + [6(377)(0.05)]^2}} = 1.26 \text{ A}$$

$$I_{12} = \frac{V_{12}}{Z_{12}} = \frac{68}{\sqrt{10^2 + [12(377)(0.05)]^2}} = 0.30 \text{ A}$$

Twelve-Pulse Rectifiers

The three-phase six-pulse bridge rectifier shows a marked improvement in the quality of the DC output over that of the single-phase rectifier. Harmonics of the output voltage are small and at frequencies that are multiples of 6 times the source frequency. Further reduction in output harmonics can be accomplished by using two six-pulse bridges as shown in Fig. 4-22*a*. This configuration is called a 12-pulse converter.

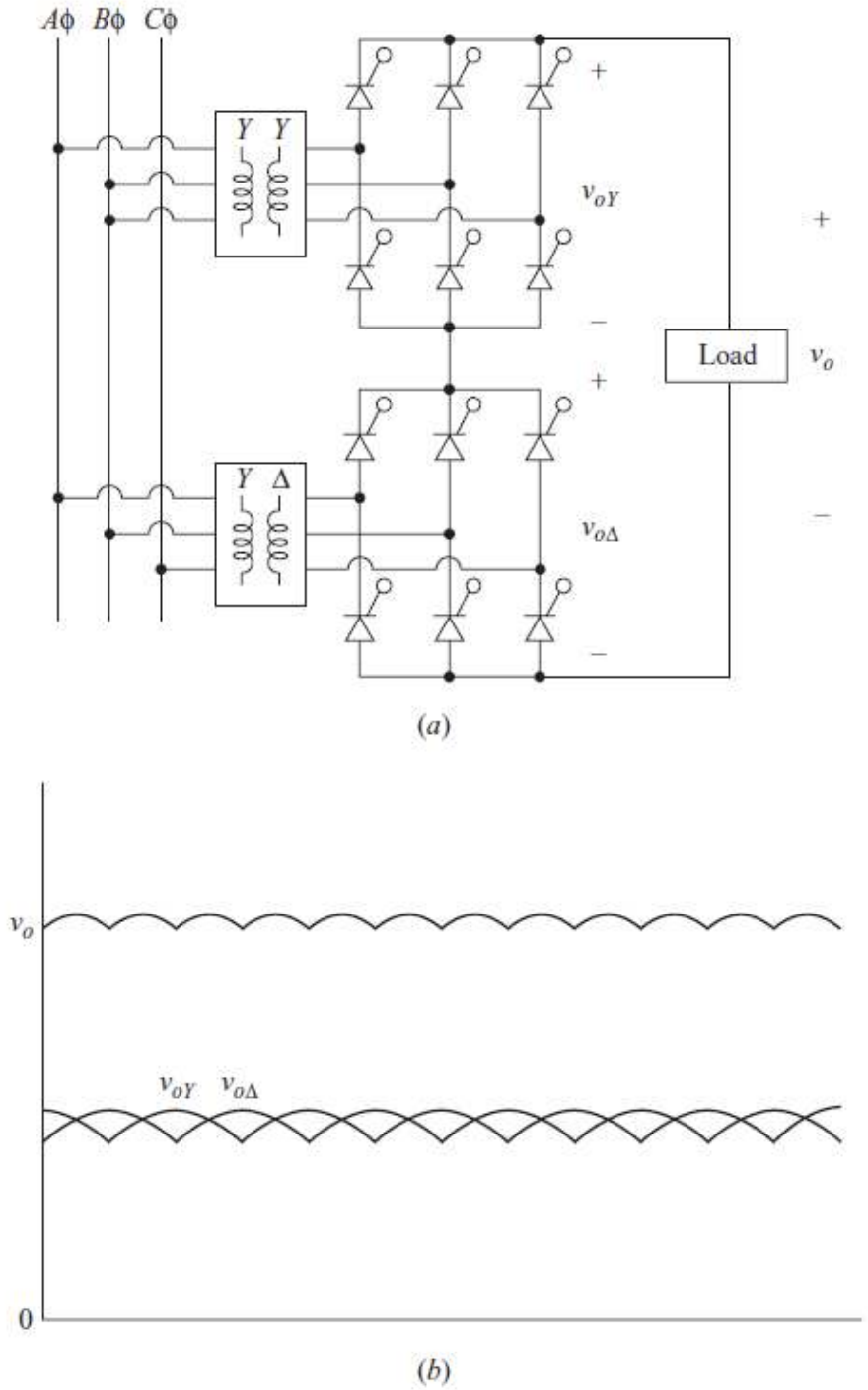


Figure 4-22 (*a*) A 12-pulse three-phase rectifier; (*b*) Output voltage for $\alpha = 0$.

One of the bridges is supplied through a *Y-Y* connected transformer, and the other is supplied through a $Y - \Delta$ (or $\Delta - Y$) transformer as shown. The purpose of the $Y - \Delta$ transformer connection is to introduce a $30°$ phase shift between the source and the bridge. This results in inputs to the two

bridges which are 30° apart. The two bridge outputs are similar, but also shifted by 30°. The overall output voltage is the sum of the two bridge outputs. The delay angles for the bridges are typically the same. The dc output is the sum of the DC output of each bridge

$$V_o = V_{o,Y} + V_{o,\Delta} = \frac{3V_{m,L-L}}{\pi}\cos\alpha + \frac{3V_{m,L-L}}{\pi}\cos\alpha = \frac{6V_{m,L-L}}{\pi}\cos\alpha \quad (4\text{-}48)$$

The peak output of the 12-pulse converter occurs midway between alternate peaks of the 6-pulse converters. Adding the voltages at that point for $\alpha = 0$ gives:

$$V_{o,\text{peak}} = 2V_{m,L-L}\cos(15°) = 1.932\ V_{m,L-L} \quad (4\text{-}49)$$

Figure 4-22*b* shows the voltages for $\alpha = 0$.
Since a transition between conducting thyristors occurs every 30°, there are a total of 12 such transitions for each period of the ac source. The output has harmonic frequencies that are multiples of 12 times the source frequency ($12k, k = 1, 2, 3, \ldots$). Filtering to produce a relatively pure dc output is less costly than that required for the 6-pulse rectifier.
Another advantage of using a 12-pulse converter rather than a 6-pulse converter is the reduced harmonics that occur in the ac system. The current in the AC lines supplying the *Y-Y* transformer is represented by the Fourier series:

$$i_Y(t) = \frac{2\sqrt{3}}{\pi}I_o\left(\cos\omega_0 t - \frac{1}{5}\cos 5\omega_0 t + \frac{1}{7}\cos 7\omega_0 t - \frac{1}{11}\cos 11\omega_0 t + \frac{1}{13}\cos 13\omega_0 t - \cdots\right) \quad (4\text{-}50)$$

The current in the ac lines supplying the $Y-\Delta$ transformer is represented by the Fourier series:

$$i_\Delta(t) = \frac{2\sqrt{3}}{\pi}I_o\left(\cos\omega_0 t + \frac{1}{5}\cos 5\omega_0 t - \frac{1}{7}\cos 7\omega_0 t - \frac{1}{11}\cos 11\omega_0 t + \frac{1}{13}\cos 13\omega_0 t + \cdots\right) \quad (4\text{-}51)$$

The Fourier series for the two currents are similar, but some terms have opposite algebraic signs. The ac system current, which is the sum of those transformer currents, has the Fourier series:

$$i_{ac}(t) = i_Y(t) + i_\Delta(t)$$
$$= \frac{4\sqrt{3}}{\pi} I_o \left(\cos \omega_0 t - \frac{1}{11} \cos 11 \omega_0 t + \frac{1}{13} \cos 13 \omega_0 t \cdots \right) \quad (4\text{-}52)$$

Thus, some of the harmonics on the ac side are canceled by using the 12-pulse scheme rather than the 6-pulse scheme. The harmonics that remain in the ac system are of order $12k \pm 1$. Cancellation of harmonics $6(2n - 1) \pm 1$ has resulted from this transformer and converter configuration. This principle can be expanded to arrangements of higher pulse numbers by incorporating increased numbers of 6-pulse converters with transformers that have the appropriate phase shifts. The characteristic ac harmonics of a p-pulse converter will be $\pm 1, k = 1, 2, 3, \ldots$. Power system converters have a practical limitation of 12 pulses because of the large expense of producing high-voltage transformers with the appropriate phase shifts. However, lower-voltage industrial systems commonly have converters with up to 48 pulses.

The Three-Phase Converter Operating as an Inverter

The above discussion focused on circuits operating as rectifiers, meaning that the power flow is from the ac side of the converter to the dc side. It is also possible for the three-phase bridge to operate as an inverter, having power flow from the DC side to the ac side. A circuit that enables the converter to operate as an inverter is shown in Fig. 4-23*a*.

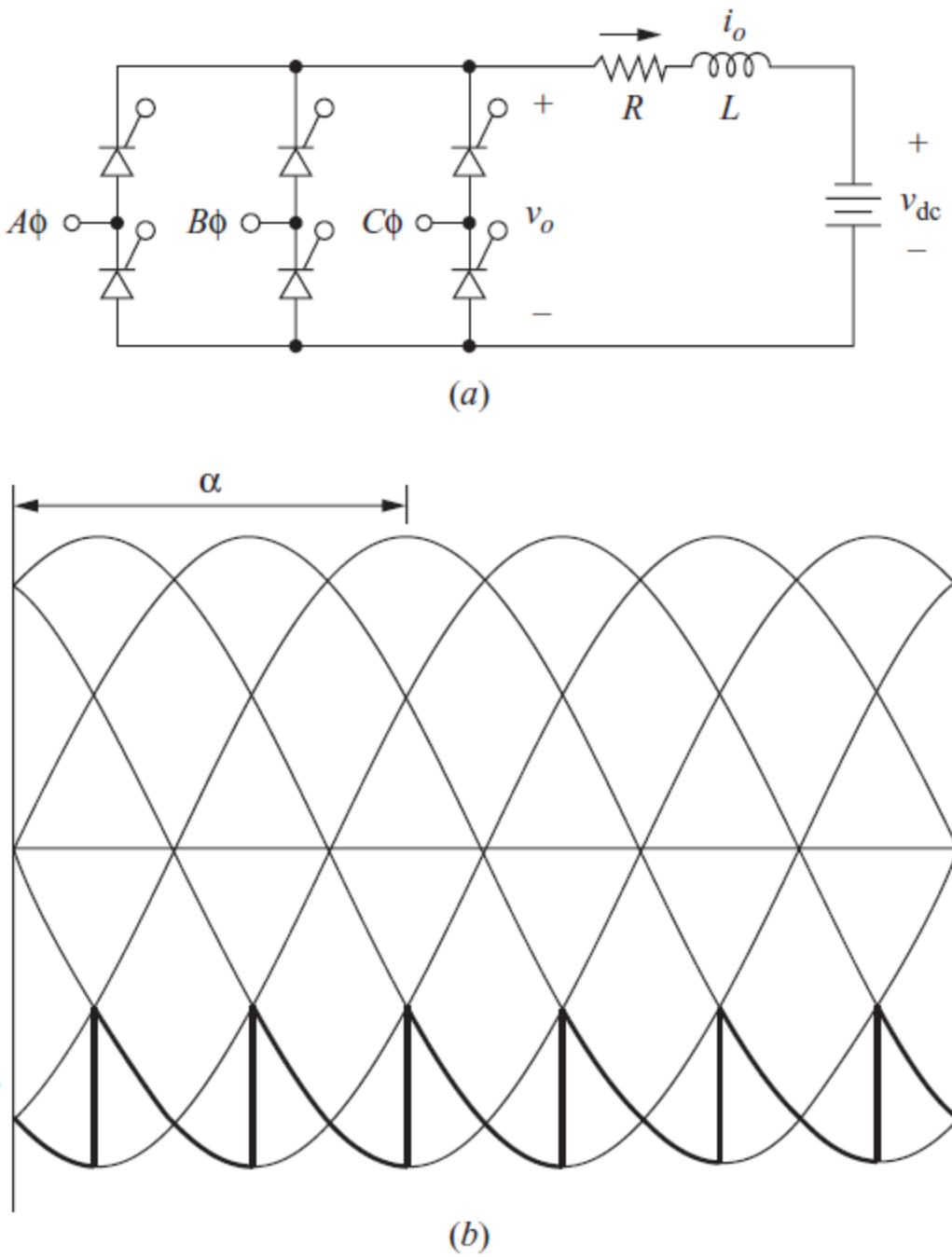


Figure 4-23 (*a*) Six-pulse three-phase converter operating as an inverter; (*b*) Bridge output voltage for $\alpha = 150°$.

Power is supplied by the DC source, and power is absorbed by the converter and transferred to the ac system. The analysis of the three-phase inverter is similar to that of the single-phase case.

The DC current must be in the direction shown because of the SCRs in the bridge. For power to be absorbed by the bridge and transferred to the ac

system, the bridge output voltage must be negative. Equation (4-47) applies, so a delay angle larger than 90° results in a negative bridge output voltage.

$$\begin{aligned} 0 < \alpha < 90^\circ \quad & V_o > 0 \quad \rightarrow \quad \text{rectifier operation} \\ 90^\circ < \alpha < 180^\circ \quad & V_o < 0 \quad \rightarrow \quad \text{inverter operation} \end{aligned} \tag{4-53}$$

The output voltage waveform for $\alpha = 150^\circ$ and continuous load current is shown in Fig. 4-23b.

Example 4-14(Three-Phase Bridge Operating as an Inverter)
The six-pulse converter of Fig. 4-23*a* has a delay angle __120_. The three-phase ac system is 4160 V rms line-to-line. The DC source is 3000 V, $R =$ 2 Ω, and L is large enough to consider the current to be purely DC.
(a) Determine the power transferred to the AC source from the dc source.
(*b*) Determine the value of L such that the peak-to-peak variation in load current is 10 percent of the average load current.

■ **Solution**

(a) The dc output voltage of the bridge is computed from Eq. (4-47) as:

$$V_o = \frac{3V_{m,L-L}}{\pi}\cos\alpha = \frac{3\sqrt{2}(4160)}{\pi}\cos(120^\circ) = -2809 \text{ V}$$

The average output current is

$$I_o = \frac{V_o + V_{\text{dc}}}{R} = \frac{-2809 + 3000}{2} = 95.5 \text{ A}$$

The power absorbed by the bridge and transferred back to the ac system is

$$P_{\text{ac}} = -I_o V_o = (-95.5)(-2809) = 268.3 \text{ kW}$$

Power supplied by the dc source is

$$P_{\text{dc}} = I_o V_{\text{dc}} = (95.5)(3000) = 286.5 \text{ kW}$$

Power absorbed by the resistance is

$$P_R = I_{\text{rms}}^2 R \approx I_o^2 R = (95.5)^2(2) = 18.2 \text{ kW}$$

(b) Variation in load current is due to the ac terms in the Fourier series. The load current amplitudes for each of the ac terms is

$$I_n = \frac{V_n}{Z_n}$$

Where V_n can be estimated from the graph of Fig. 4-21 and:

$$Z_n = |R + jn\omega_0 L|$$

Since the decreasing amplitude of the voltage terms and the increasing magnitude of the impedance both contribute to diminishing ac currents as n increases, the peak-to-peak current variation will be estimated from the first ac term. For $n = 6$, V_n/V_m is estimated from Fig. 4-21 as 0.28, making $V6 = 0.28(4160)\sqrt{2} = 1650\ V$. The peak-to-peak variation of 10 percent corresponds to a zero-to-peak amplitude of $(0.05)(95.5) = 4.8\ A$. The required load impedance for $n = 6$ is then:

$$Z_6 = \frac{V_6}{I_6} = \frac{1650\ \text{V}}{4.8\ \text{A}} = 343\ \Omega$$

The 2 Ω resistor is insignificant compared to the total 343 Ω required impedance, so $Z_6 \approx 6\ \omega_o L$. Solving for L,

$$L \approx \frac{Z_6}{6\omega_0} = \frac{343}{6(377)} = 0.15\ \text{H}$$

4.6 DC Power Transmission

The controlled 12-pulse converter of Fig. 4-22*a* is the basic element for dc power transmission. DC transmission lines are commonly used for transmission of electric power over very long distances. Examples include the Pacific Intertie; the Square Butte Project from Center, North Dakota, to Duluth, Minnesota; and the Cross Channel Link under the English Channel between England and France. Modern DC lines use SCRs in the converters, while very old converters used mercury-arc rectifiers.

Advantages of DC power transmission include the following:

The inductance of the transmission line has zero impedance to dc, whereas the inductive impedance for lines in an AC system is relatively large.

2. The capacitance that exists between conductors is an open circuit for dc. For ac transmission lines, the capacitive reactance provides a path for current, resulting in additional I^2R losses in the line. In applications where the conductors are close together, the capacitive reactance can be a

significant problem for ac transmission lines, whereas it has no effect on dc lines.

3. There are two conductors required for dc transmission rather than three for conventional three-phase power transmission. (There will likely be an additional ground conductor in both DC and AC systems.)

4. Transmission towers are smaller for dc than ac because of only two conductors, and right-of-way requirements are less.

5. Power flow in a dc transmission line is controllable by adjustment of the delay angles at the terminals. In an ac system, power flow over a given transmission line is not controllable, being a function of system generation and load.

6. Power flow can be modulated during disturbances on one of the ac systems, resulting in increased system stability.

7. The two AC systems that are connected by the DC line do not need to be in synchronization. Furthermore, the two ac systems do not need to be of the same frequency. A 50-Hz system can be connected to a 60-Hz system via a DC link.

The disadvantage of dc power transmission is that a costly ac-dc converter, filters, and control system are required at each end of the line to interface with the AC system.

Figure 4-24a shows a simplified scheme for dc power transmission using six-pulse converters at each terminal.

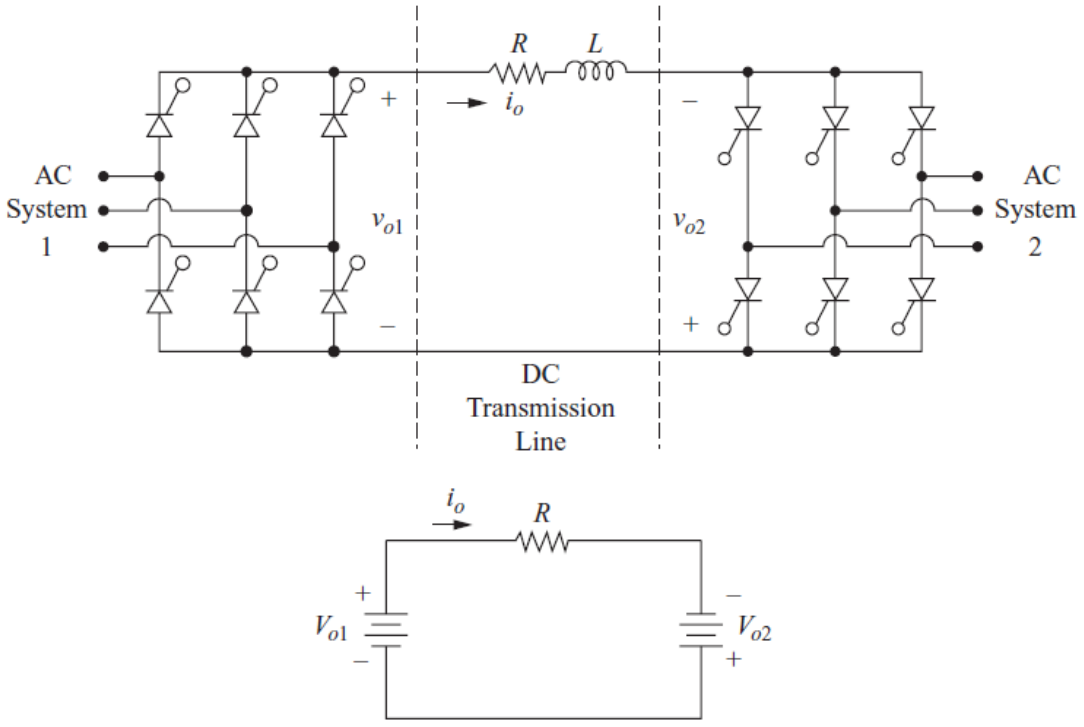


Figure 4-24 (*a*) An elementary dc transmission system; (*b*) Equivalent circuit.

The two AC systems each have their own generators, and the purpose of the dc line is to enable power to be interchanged between the AC systems. The directions of the SCRs are such that the current i_o will be positive as shown in the line.

In this scheme, one converter operates as a rectifier (power flow from AC to DC), and the other terminal operates as an inverter (power flow from DC to AC). Either terminal can operate as a rectifier or inverter, with the delay angle determining the mode of operation. By adjusting the delay angle at each terminal, power flow is controlled between the two AC systems via the DC link.

The inductance in the dc line is the line inductance plus an extra series inductor to filter harmonic currents. The resistance is that of the dc line conductors.

For analysis purposes, the current in the dc line may be considered to be a ripple-free DC current.

Voltages at the terminals of the converters V_{o1} and V_{o2} are positive as shown for α between 0 and 90° and negative for α between 90° and 180°. The converter supplying power will operate with a positive voltage while the converter absorbing power will have a negative voltage.

With converter 1 in Fig. 4-24a operating as a rectifier and converter 2 operating as an inverter, the equivalent circuit for power computations is shown in Fig. 4-24b. The current is assumed to be ripple-free, enabling only the DC component of the Fourier series to be relevant. The DC current is:

$$I_o = \frac{V_{o1} + V_{o2}}{R} \qquad (4\text{-}54)$$

Where:

$$V_{o1} = \frac{3V_{m1,L-L}}{\pi} \cos \alpha_1$$

$$V_{o2} = \frac{3V_{m2,L-L}}{\pi} \cos \alpha_2 \qquad (4\text{-}55)$$

Power supplied by the converter at terminal 1 is:

$$P_1 = V_{o1} I_o \qquad (4\text{-}56)$$

Power supplied by the converter at terminal 2 is:

$$P_2 = V_{o2} I_o \qquad (4\text{-}57)$$

Example 4-15(DC Power Transmission)
For the elementary dc transmission line represented in Fig. 4-24*a*, the ac voltage to each of the bridges is $230\ kV\ rms$ line to line. The total line resistance is $10\ \Omega$, and the inductance is large enough to consider the DC current to be ripple-free. The objective is to transmit 100 MW to AC system 2 from AC system 1 over the DC line. Design a set of operating parameters to accomplish this objective. Determine the required current-carrying capacity of the DC line, and compute the power loss in the line.

■ **Solution**

The relationships that are required are from Eqs. (4-54) to (4-57), where:

$$P_2 = I_o V_{o2} = -100 \text{ MW} \qquad (100 \text{ MW absorbed})$$

The maximum dc voltage that is obtainable from each converter is, for $\alpha = 0$ in Eq. (4-47),

$$V_{o,\,\max} = \frac{3V_{m,L-L}}{\pi} = \frac{3\sqrt{2}\,(230 \text{ kV})}{\pi} = 310.6 \text{ kV}$$

The dc output voltages of the converters must have magnitudes less than 310.6 kV, so a voltage of $-200\ kV$ is arbitrarily selected for converter 2. This voltage must be negative because power must be absorbed at converter 2. The delay angle at converter 2 is then computed from Eq. (4-47).

$$V_{o2} = \frac{3V_{m,L-L}}{\pi} \cos \alpha_2 = (310.6 \text{ kV}) \cos \alpha_2 = -200 \text{ kV}$$

Solving for α_2,

$$\alpha_2 = \cos^{-1}\left(\frac{-200 \text{ kV}}{310.6 \text{ kV}}\right) = 130°$$

The DC current required to deliver 100 MW to converter 2 is then

$$I_o = \frac{100 \text{ MW}}{200 \text{ kV}} = 500 \text{ A}$$

Which is the required current-carrying capacity of the line.
The required dc output voltage at converter 1 is computed as

$$V_{o1} = -V_{o2} + I_o R = 200 \text{ kV} + (500)(10) = 205 \text{ kV}$$

The required delay angle at converter 1 is computed from Eq. (4-47).

$$\alpha_1 = \cos^{-1} \frac{205 \text{ kV}}{310.6 \text{ kV}} = 48.7°$$

Power loss in the line is$I_{rms}^2 R$, where $I_{rms} \approx I_o$ because the ac components of line current are filtered by the inductor. Line loss is:

$$P_{\text{loss}} = I_{\text{rms}}^2 R \approx (500)^2 (10) = 2.5 \text{ MW}$$

Note that the power supplied at converter 1 is

$$P_1 = V_{\text{dc1}} I_o = (205 \text{ kV})(500 \text{ A}) = 102.5 \text{ MW}$$

Which is the total power absorbed by the other converter and the line resistance.

Certainly other combinations of voltages and current will meet the design objectives, as long as the DC voltages are less than the maximum possible output voltage and the line and converter equipment can carry the current. A better design might have higher voltages and a lower current to reduce power loss in the line. That is one reason for using 12-pulse converters and bipolar operation, as discussed next.

A more common DC transmission line has a 12-pulse converter at each terminal.

This suppresses some of the harmonics and reduces filtering requirements. Moreover, a pair of 12-pulse converters at each terminal provides bipolar operation. One of the lines is energized at $-V_{dc}$ and the other is energized at$+V_{dc}$. In emergency situations, one pole of the line can operate without the other pole, with current returning through the ground path. Figure 4-25 shows a bipolar scheme for dc power transmission.

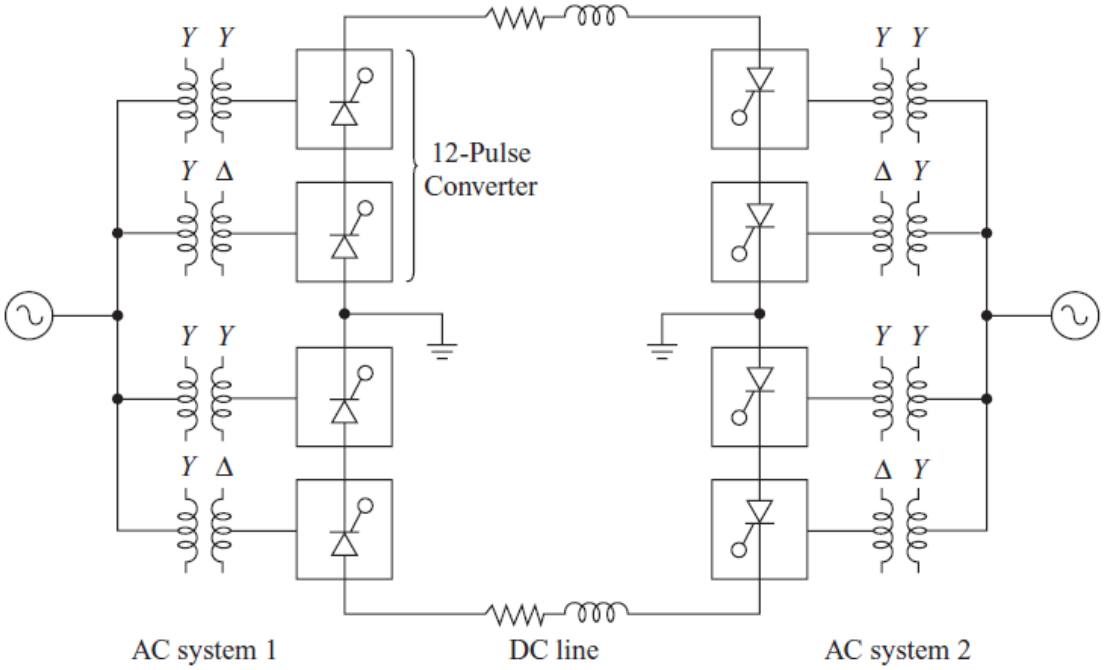


Figure 4-25 A dc transmission system with two 12-pulse converters at each terminal.

4.7 Commutation: The Effect Of Source Inductance

Single-Phase Bridge Rectifier

An uncontrolled single-phase bridge rectifier with a source inductance $L_S$and an inductive load is shown in Fig. 4-26*a*.

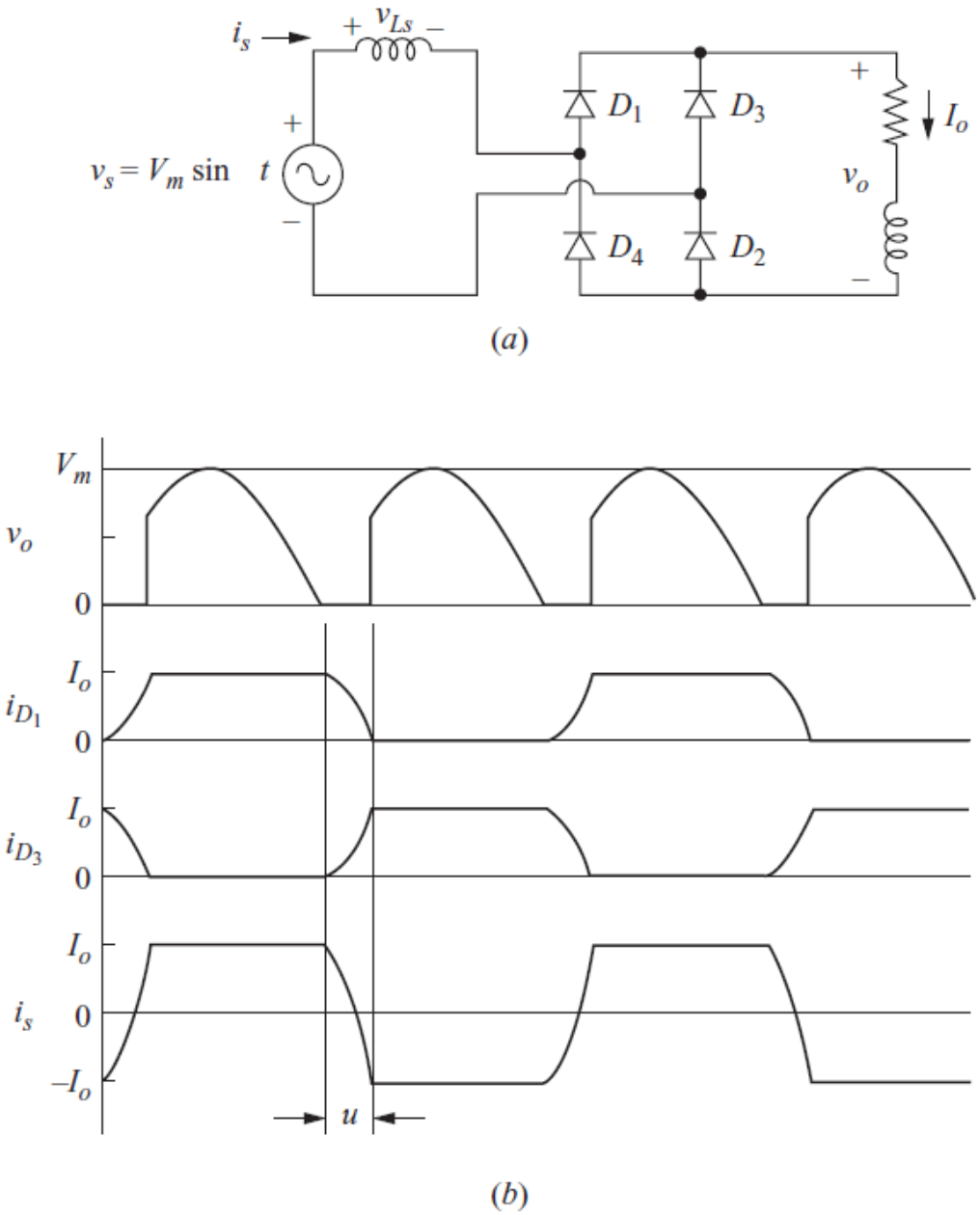


Figure 4-26 Commutation for the single-phase rectifier (*a*) circuit with source inductance L_S; (*b*) voltage and current waveforms.

When the source changes polarity, source current cannot change instantaneously, and current must be transferred gradually from one diode pair to the other over a commutation interval u, as shown in Fig. 4-26*b*. Recall from Chap. 3 that commutation is the process of transferring the load current from one diode to another or, in this case, one diode pair to the other. (See Sec. 3.11.) During commutation, all four diodes are on, and the voltage across $L_S$is the source voltage $V_m \; sin \; (\omega t)$.

Assume that the load current is a constant I_o. The current in L_S and the source during the commutation from $D_1 - D_2$ to $D_3 - D_4$ starts at $+ I_o$ and goes to $-I_o$. This commutation interval starts when the source changes polarity at $\omega t = \pi$ as is expressed in:

$$i_s(\omega t) = \frac{1}{\omega L_s} \int_{\pi}^{\omega t} V_m \sin(\omega t)\, d(\omega t) + I_o$$

Evaluating,

$$i_s(\omega t) = -\frac{V_m}{\omega L_s}(1 + \cos \omega t) + I_o \qquad (4\text{-}58)$$

When commutation is complete at $\omega t = \pi + u$,

$$i(\pi + u) = -I_o = -\frac{V_m}{\omega L_S}[1 + \cos(\pi + u)] + I_o \qquad (4\text{-}59)$$

Solving for the commutation angle u,

$$u = \cos^{-1}\left(1 - \frac{2I_o \omega L_S}{V_m}\right) = \cos^{-1}\left(1 - \frac{2I_o X_S}{V_m}\right) \qquad (4\text{-}60)$$

Where $X_S = \omega\, L_S$ is the reactance of the source. Figure 4-26b shows the effect of the source reactance on the load current and voltage.
Average load voltage is:

$$V_o = \frac{1}{\pi} \int_{u}^{\pi} V_m \sin(\omega t)\, d(\omega t) = \frac{V_m}{\pi}(1 + \cos u)$$

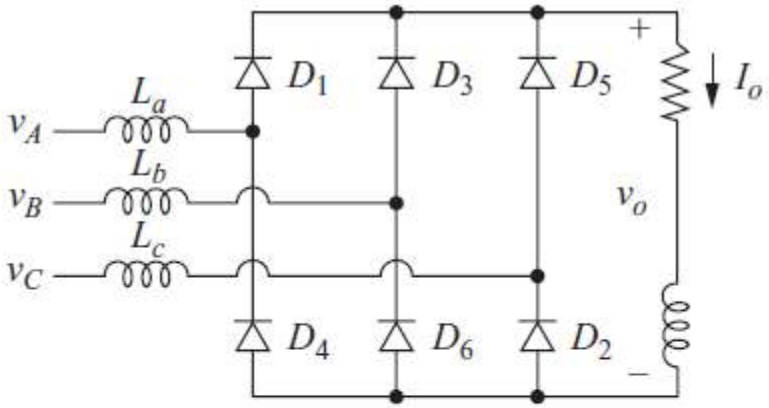
Using u from Eq. (4-60),

$$\boxed{V_o = \frac{2V_m}{\pi}\left(1 - \frac{I_o X_s}{V_m}\right)}$$

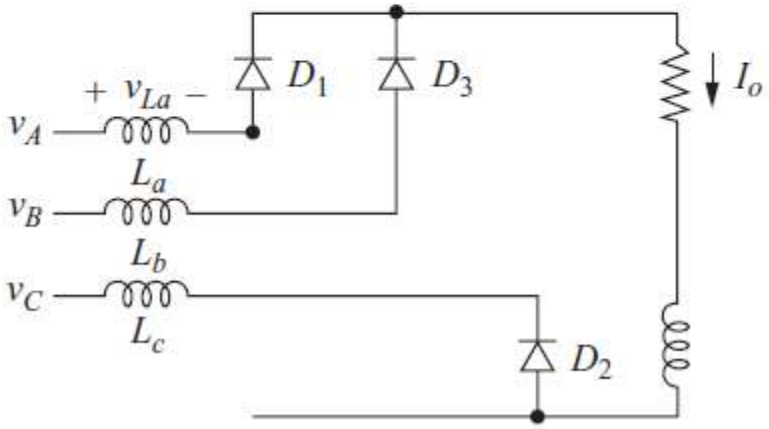
Thus, source inductance lowers the average output voltage of full-wave rectifiers.

Three-Phase Rectifier

For the uncontrolled three-phase bridge rectifier with source reactance (Fig. 4-27a), assume that diodes D_1 and D_2 are conducting and the load current is a constant I_o.



(*a*)



(*b*)

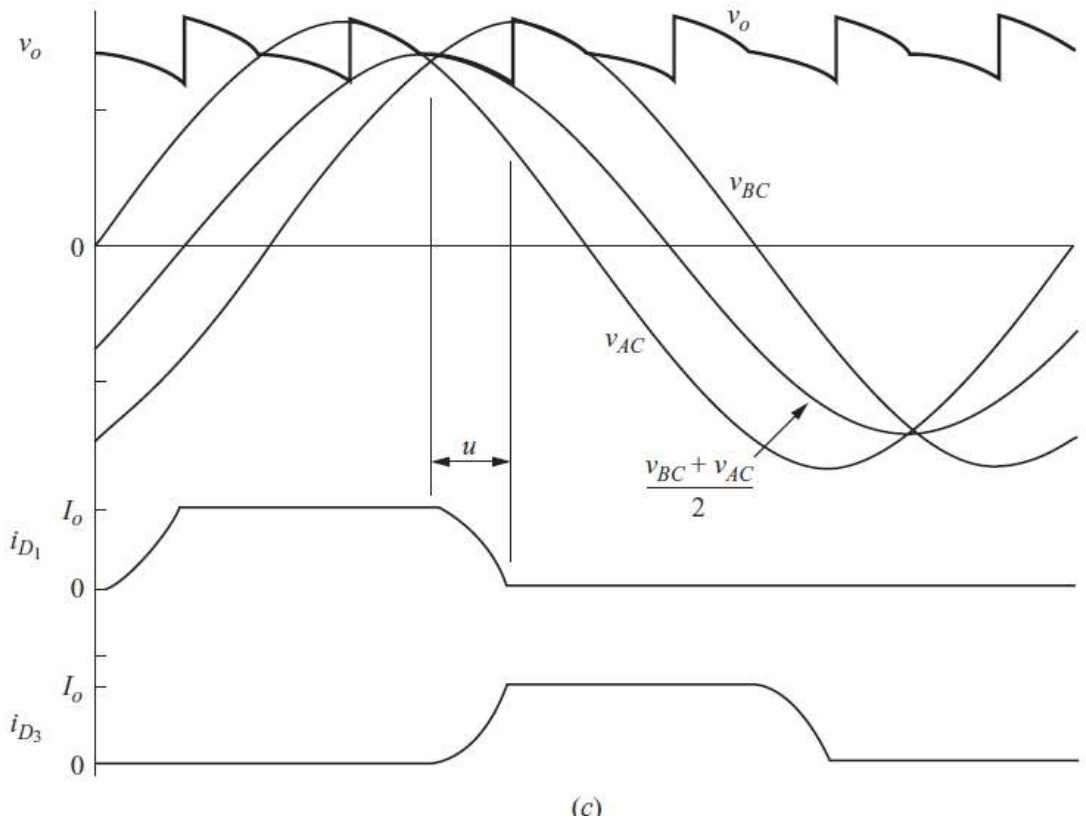


Figure 4-27 Commutation for the three-phase rectifier. (*a*) Circuit; (*b*) Circuit during commutation from D_1 to D_3; (*c*) Output voltage and diode currents.

The next transition has load current transferred from D_1 to D_3 in the top half of the bridge. The equivalent circuit during commutation from D_1 to D_3 is shown in Fig. 4-27*b*. The voltage across L_a is:

$$v_{La} = \frac{v_{AB}}{2} = \frac{V_{m,L-L}}{2}\sin(\omega t) \tag{4-62}$$

Current in L_a starts at I_o and decreases to zero in the commutation interval,

$$i_{La}(\pi + u) = 0 = \frac{1}{\omega L_a}\int_{\pi}^{\pi+u} \frac{V_{m,L-L}}{2}\sin(\omega t)\,d(\omega t) + I_o \tag{4-63}$$

Solving for u,

$$u = \cos^{-1}\left(1 - \frac{2\omega L_a I_o}{V_{m,L-L}}\right) = \cos^{-1}\left(1 - \frac{2X_s I_o}{V_{m,L-L}}\right) \tag{4-64}$$

During the commutation interval from D_1 to D_3, the converter output voltage is

$$v_o = \frac{v_{bc} + v_{ac}}{2} \qquad (4\text{-}65)$$

Output voltage and diode currents are shown in Fig. 4-27c. Average output voltage for the three-phase converter with a nonideal source is

$$V_o = \frac{3V_{m,L-L}}{\pi}\left(1 - \frac{X_s I_o}{V_{m,L-L}}\right) \qquad (4\text{-}66)$$

Therefore, source inductance lowers the average output voltage of three-phase rectifiers.

4.8 Summary

• Single-phase full-wave rectifiers can be of the bridge or center-tapped transformer types.
• The average source current for single-phase full-wave rectifiers is zero.
• The Fourier series method can be used to analyze load currents.
  - A large inductor in series with a load resistor produces a load current that is essentially dc.
• A filter capacitor on the output of a rectifier can produce an output voltage that is nearly dc. An LC output filter can further improve the quality of the dc output and reduce the peak current in the diodes.
• Switches such as SCRs can be used to control the output of a single-phase or three-phase rectifier.
• Under certain circumstances, controlled converters can be operated as inverters.
• The 6-pulse three-phase rectifiers have 6 diodes or SCRs, and 12-pulse rectifiers have 12 diodes or SCRs.
• Three-phase bridge rectifiers produce an output that is inherently like dc.
• DC power transmission has a three-phase converter at each end of a dc line. One converter is operated as a rectifier and the other is operated as a converter.
• Source inductance reduces the dc output of a single-phase or three-phase rectifier.